Learning to Count: Discovering Enumerative Combinatorics

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This book is intended to be a problem sequence for a one-semester graduate enumerative combinatorics course that utilizes an inquiry-based learning (IBL) approach. You may not modify the content of this book without expressed consent of the author.

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> The mathematician does not study pure mathematics because it is useful; he studies it because he delights in it, and he delights in it because it is beautiful.
>
> Henri Poincaré, mathematician & physicist

Chapter 1

Introduction

1.1 What is This Course All About?

This course focuses on enumerative combinatorics with an introduction to generating function techniques.

This course will use an inquiry-based learning (IBL) to learning combinatorics. Rather than spending most of class listening to lectures, you will spend substantial time exploring problems, developing arguments, presenting solutions, and discussing the mathematics with your classmates.

Outside of class, you may use whatever resources help you learn—including books, websites, classmates, and AI tools such as ChatGPT. The important distinction is between **finding a solution** and **understanding the mathematics**. When you come to class, you should be prepared to explain, defend, critique, and adapt the ideas you encountered.

Because outside resources are readily available, homework and preparation will be relatively low-stakes. Your understanding will be assessed primarily through your mathematical contributions in class and regular short, individual, no-resource assessments. In short:

Use whatever helps you learn. Be prepared to demonstrate that you learned it.

1.2 Inquiry-Based Learning

Mathematics is not about calculations, but ideas. My goal as a teacher is to provide students with the opportunity to grapple with these ideas and to be immersed in the process of mathematical discovery. Repeatedly engaging in this process hones the mind and develops mental maturity marked by clear and rigorous thinking. Like music and art, mathematics provides an opportunity for enrichment, experiencing beauty, elegance, and aesthetic value. The medium of a painter is color and shape, whereas the medium of a mathematician is abstract thought. The creative aspect of mathematics is what captivates

me and fuels my motivation to keep learning and exploring.

While the content we teach our students is important, it is not enough. An education must prepare individuals to ask and explore questions in contexts that do not yet exist and to be able to tackle problems they have never encountered. It is important that we put these issues front and center and place an explicit focus on students producing, rather than consuming, knowledge. If we truly want our students to be independent, inquisitive, and persistent, then we need to provide them with the means to acquire these skills. Their viability as a professional in the modern workforce depends on their ability to embrace this mindset.

When I started teaching, I mimicked the experiences I had as a student. Because it was all I knew, I lectured. By standard metrics, this seemed to work out just fine. Glowing student and peer evaluations, as well as reoccurring teaching awards, indicated that I was effectively doing my job. People consistently told me that I was an excellent teacher. However, two observations made me reconsider how well I was really doing. Namely, many of my students seemed to depend on me to be successful, and second, they retained only some of what I had taught them. In the words of Dylan Retsek:

"Things my students claim that I taught them masterfully, they don't know."

Inspired by a desire to address these concerns, I began transitioning away from direct instruction towards a more student-centered approach. The goals and philosophy behind inquiry-based learning (IBL) resonate deeply with my ideals, which is why I have embraced this paradigm. According to the Academy of Inquiry-Based Learning, IBL is a method of teaching that engages students in sense-making activities. Students are given tasks requiring them to solve problems, conjecture, experiment, explore, create, and communicate—all those wonderful skills and habits of mind that mathematicians engage in regularly.

In an IBL course, instructor and students have joint responsibility for the depth and progress of the course. While effective IBL courses come in a variety of forms, they all possess a few essential ingredients. According to Laursen and Rasmussen (2019), the Four Pillars of IBL are:

- Students engage deeply with coherent and meaningful mathematical tasks.
- Students collaboratively process mathematical ideas.
- Instructors inquire into student thinking.
- Instructors foster equity in their design and facilitation choices.

This book can only address the first pillar while it is the responsibility of your instructor and class to develop a culture that provides an adequate environment for the remaining pillars to take root.

Just like learning to play an instrument or sport, you will have to learn new skills and ideas. Along this journey, you should expect a cycle of victory and defeat, experiencing a full range of emotions. Sometimes you will feel exhilarated, other times you might

be seemingly paralyzed by extreme confusion. You will experience struggle and failure before you experience understanding. This is part of the normal learning process. If you are doing things well, you should be confused on a regular basis. Productive struggle and mistakes provide opportunities for growth. As the author of this text, I am here to guide and challenge you, but I cannot do the learning for you, just as a music teacher cannot move your fingers and your heart for you.

You could view this book as mountaineering guidebook. I have provided a list of mountains to summit, sometimes indicating which trailhead to start at or which trail to follow. There will always be multiple routes to top, some more challenging than others. Some summits you will attain quickly and easily, others might require a multi-day expedition. Oftentimes, your journey will be laced with false summits. Some summits will be obscured by clouds. Sometimes you will have to wait out a storm, perhaps turning around and attempting another route, or even attempting to summit on a different day after the weather has cleared. The strength, fitness, and endurance you gain along the way will allow you to take on more and more challenging, and often beautiful, terrain. Do not forget to take in the view from the top! The joy you feel from overcoming obstacles and reaching each summit under your own will and power has the potential to be life changing. But make no mistake, the journey is vastly more important than the destinations.

> Don't fear failure. Not failure, but low aim, is the crime. In great attempts it is glorious even to fail.
>
> Attributed to Bruce Lee, martial artist & actor

Chapter 2

Preliminaries

2.1 Combinatorial Structures

One of our objectives is to study combinatorial structures that arise in mathematics, with an eye towards enumeration and classification. We will begin by investigating several sets of discrete mathematical objects and exploring the connections among them. In no particular order, here are some combinatorial species[1] we may encounter throughout the semester:

Binary String	Pixel Strip	Lattice Path
Starfolks Path	Compositions	Pebble Piles
Domino Stacking	Domino Tiling	Permutation
Nonattacking Rooks	Sparse Strip	Odd Composition
Binary Tree	Increasing Tree	Triangulated Polygon
Ballot Path	Noncrossing Partition	Parking Function
Unlabeled Tree	Set Partition	Integer Partition
Dodger Rooks	Cayley Permutation	Fubini Arrangement
Fraenkel Tree	Knuth Rooks	Penrose Doodle

Note that I invented some of the names mentioned above. Figure 2.1 provides a few examples of individuals one can find among the species mentioned above.

Below are some questions we should ponder when exploring a new species:

- What patterns do we see?
- How do they "grow"?
- What is a sensible definition of the species? How does your definition tie in to the **index** n and the **rank** r?
- Can you provide a generic example with index $n = 10$? For your example, what is

[1] In this context, I am using the word "species" in a colloquial sense. Later in the semester, this word will have a more precise mathematical meaning.

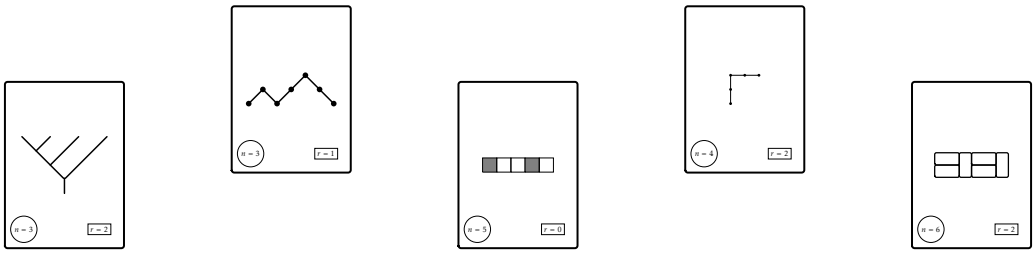


Figure 2.1: Examples of combinatorial species.

r?

- How many objects are there of index n? Is there a nice formula?
- How many objects are there of index n and rank r? Is there a nice formula?
- For a fixed index n how do you generate all the possibilities? What is the simplest/most efficient way to generate them?
- How does this species relate to other species we have studied?

You may not be able to answer some of these questions immediately, and that's okay. The goals here are to have a variety of examples at our finger tips and to start a taxonomy of combinatorial species.

Problem 2.1. For each of the species you encountered or were assigned, complete the following.

(a) List the name of the species.

(b) Provide a working definition for the species. Your definition should include mention of the index n and the rank r, including the domain for n (and possibly r). For example, your definition might start as, "A *dinglehopper* of index $n \in \mathbb{N} \cup \{0\}$ and rank r is…"

(c) Provide a generic example for index $n = 10$. For your example, specify the rank r. Your example should illustrate most or all of the key features and not be too "boring."

(d) Provide a table of values that enumerates the number of structues of index n for small n. Your table should start at either $n = 0$ or $n = 1$ depending on the species and it should look something like the following:

n	0	1	2	3	4	5
count	1	1	2	3	5	8

(e) If you can, write down a formula (either a closed form or recursive) that counts the number of structures of index n. Justify your answer. Please do not look this up or ask AI. If you cannot do it at this time, I would prefer we leave it blank!

(f) Now, provide a table of values that enumerates the number of structures of index n and rank r for small values of n and r. Your table should start at either $n = 0$ or $n = 1$ and $r = 0$ or $r = 1$ depending on the species, with n labeling the rows and r labeling the columns. Table 2.1 illustrates the requested format.

$n\backslash r$	0	1	2	3	4	5
0	1					
1	1	1				
2	1	2	1			
3	1	3	3	1		
4	1	4	6	4	1	
5	1	5	10	10	5	1

Table 2.1: Sample counts of structures organized by index n and rank r.

(g) Ignoring the row labels, what happens if you sum the enteries in a row of your table in part (f)?

(h) If you can, write down a formula (either a closed form or recursive) that counts the number of structures of index n and rank r. Justify your answer. Please do not look this up or ask AI. If you cannot do it at this time, I would prefer we leave it blank!

(i) Are there any other species you have already encountered that seems related to this species?

2.2 First Principles

Let's establish some notation and make sure we are on the same page regarding some basic counting principles.

In this book, the set of **natural numbers** is given by $\mathbb{N} := \{1, 2, 3, \ldots\}$. Sometimes mathematicians include 0 in the natural numbers, but that will not be our convention. For $n \in \mathbb{N} \cup \{0\}$, the following notation is useful:

$$[n] := \begin{cases} \{1, 2, \ldots, n\}, & \text{if } n \in \mathbb{N} \\ \varnothing, & \text{if } n = 0. \end{cases}$$

The **cardinality** of a set A, denoted by $|A|$, is the number of elements in A. If A and B are sets, the **Cartesian product** of A and B, denoted $A \times B$ (read as "A times B" or "A cross B"), is the set of all **ordered pairs** where the first component is from A and the second component is from B. In set-builder notation, we have

$$A \times B := \{(a, b) \mid a \in A, b \in B\}.$$

Cartesian products are named after French philosopher and mathematician René Descartes (1596–1650).

Problem 2.2. If A is a set, then what is $A \times \varnothing$ equal to?

Problem 2.3. Under what circumstances is $A \times B = B \times A$?

The Cartesian product of sets $A_1, \ldots, A_n$ is the collection of n-**tuples** given by

$$A_1 \times \cdots \times A_n := \{(a_1, \ldots, a_n) \mid a_j \in A_j \text{ for all } 1 \leq j \leq n\},$$

where A_i is referred to as the ith **factor** of the Cartesian product. As a special case, the set

$$\underbrace{A \times \cdots \times A}_{n \text{ factors}}$$

is often abbreviated as A^n.

Theorem 2.4 (Product Principle)**.** Each of the following are referred to as the **Product Principle**.

(a) If $A_1, \ldots, A_k$ are finite sets, then

$$|A_1 \times \cdots \times A_k| = |A_1| \cdots |A_k|.$$

(b) If $\mathcal{O}$ is the set of outcomes for a k-step process, where for $1 \leq i \leq k$, there are n_i choices for step i, no matter what earlier choices were made, then

$$|\mathcal{O}| = n_1 n_2 \cdots n_k.$$

The key difference between the two versions of the Product Principle is that the second version does not assume that the set of choices for step i is independent of the previous choices.

Problem 2.5. Is your answer for Problem 2.2 compatible with the Product Principle?

For $n \in \mathbb{N} \cup \{0\}$, a **string** of objects of **length** n is a linearly ordered arrangements of n objects. The objects in a string may be repeated unless specified otherwise. A string of length 0 is referred to as the **empty string**. If a string consists of letters, then it may be called a **word** (we do not require a word to be in the dictionary!). For example, $NNENEEE$ is a word of length 7 consisting of N's and E's.

For $n \in \mathbb{N} \cup \{0\}$, a **binary string** is a string consisting of 0's and 1's. Each occurrence of a 0 or 1 in a binary string is called a **bit**. The strings 010 and 100 are two different binary strings of length 3.

Problem 2.6. For $n \in \mathbb{N} \cup \{0\}$, how many binary strings of length n are there?

Problem 2.7. For $n \in \mathbb{N} \cup \{0\}$, how many subsets does a set with n elements have? You should be able to carefully justify your answer by cleverly utilizing the Product Principle.

For $n \in \mathbb{N}$, a **pebble pile configuration** of index n and rank r consists of n indistinguishable dots arranged into r nonempty vertical piles. Each pile is bottom-justified along a common horizontal baseline, with the dots in a pile stacked vertically. The piles are arranged from left to right, and their order matters. Figure 2.2 depicts a pebble pile configuration with 10 pebbles and 4 piles.

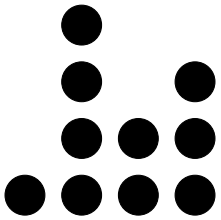


Figure 2.2: A pebble pile configuration consisting of 10 pebbles and 4 piles.

Problem 2.8. For $n \in \mathbb{N}$, how many pebble pile configurations are there consisting of n pebbles?

For $n \in \mathbb{N}$ and $1 \leq r \leq n$, a **composition** of n with r parts is an ordered list of r positive integers whose sum is n. For example, $(1, 2, 1, 3)$ is a composition of 7 with 4 parts. An arbitrary composition with r parts may be denoted as $\alpha = (\alpha_1, \ldots, \alpha_r)$.

Problem 2.9. For $n \in \mathbb{N}$, how many compositions of n are there? Is there an obvious connection between pebble pile configurations and compositions? Can you explicitly describe this connection?

Problem 2.10. Imagine we have $n \geq 1$ distinct balls labeled $1, 2, \ldots, n$ and $k \geq 1$ distinct buckets labeled $1, 2, \ldots, k$. How many ways can we distribute the n distinct balls into the k distinct buckets?

Problem 2.11. Let $n, k \in \mathbb{N}$.

(a) How many functions are there from $[n]$ to $[k]$?

(b) If $k \leq n$, how many injective functions are there from $[k]$ to $[n]$?

(c) How many bijective functions are there from $[n]$ to $[n]$?

The game of chess is played on an 8×8 grid. We will consider chessboards with arbitrary dimensions, say $n \times k$ (n rows, k columns) with $1 \leq k \leq n$. A rook is a castle-shaped piece that can move horizontally or vertically any number of squares. In a typical chess game, there are two black rooks and two white rooks, but for our purposes we will assume we have k rooks, where all the rooks are the same color. We say that an arrangement of rooks on an $n \times k$ board is **nonattacking** if no two rooks lie in the same row or column. Figure 2.3 shows one possible nonattacking rook arrangement on a 4×4 board.

Problem 2.12. For $n \in \mathbb{N}$ and $1 \leq k \leq n$, how many nonattacking rook arrangements are there on an $n \times k$ board? What about an $n \times n$ board?

Theorem 2.13 (Sum Principle)**.** If $A_1, \ldots, A_n$ are pairwise disjoint sets, then

$$|A_1 \cup \cdots \cup A_n| = |A_1| + \cdots + |A_n|.$$

The Sum Principle is especially useful when a collection of objects can be partitioned into disjoint cases: count the objects in each case, then add the resulting counts.

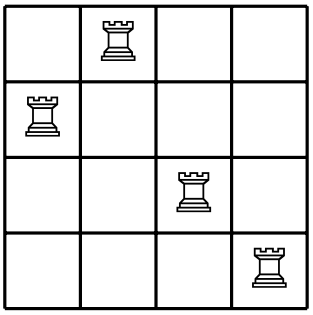


Figure 2.3: A nonattacking rook arrangement on a 4×4 board.

Loosely speaking, a sequence of numbers is defined recursively if the nth term of the sequence is defined in terms of "earlier" terms of the sequence. One famous example of a sequence being defined recursively is the **Fibonacci sequence** (f_n), which we define by $f_0 = 1$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. The equation $f_n = f_{n-1} + f_{n-2}$ is the **recurrence relation** while $f_0 = 1$ and $f_1 = 1$ are the **initial conditions**. The first few terms of the Fibonacci sequence are:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$$

It is important to emphasize that we cannot define the Fibonacci numbers using only the recurrence relation since otherwise, we would not be able to "get started" with the recurrence.

Problem 2.14. How many compositions α of n have the following properties?

(a) α has parts of size 1 and 2 only.

(b) α has only odd parts.

Problem 2.15. For $n \in \mathbb{N} \cup \{0\}$, how many binary strings of length n never have two consecutive 1's?

Problem 2.16. For $n \in \mathbb{N}$, consider a $2 \times n$ grid of squares. Using n dominoes with dimensions 1×2 squares, how many ways can we tile (i.e., perfectly cover with no overlap) the grid?

If a finite collection of sets is not pairwise disjoint, then we can still enumerate their union using the **Principle of Inclusion and Exclusion**.

Theorem 2.17 (Principle of Inclusion and Exclusion)**.** The number of elements in the union of finite sets $A_1, A_2, \ldots, A_n$ is

$$|A_1 \cup \cdots \cup A_n| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cdots \cap A_n|.$$

Problem 2.18. To ensure you understand the formula in the previous theorem, write down formulas for computing $|A \cup B|$ and $|A \cup B \cup C|$.

The next result, sometimes referred to as the **Subtraction Principle**, is simply a special case of Sum Principle.

Theorem 2.19 (Subtraction Principle)**.** If $A \subseteq U$, then $|A| = |U| - |A^c|$.

There are two use cases for the Subtraction Principle. Sometimes it might be easier to count the opposite of what you are looking to count and then to subtract from the size of the universe of discourse. The second approach involves "inflating" the universe of discourse to include additional elements, counting the larger set, and then subtracting off the extra stuff. The two approaches are really just two different perspectives of the same idea.

The **Division Principle** provides a method for ignoring "unimportant" differences when counting things. Loosely speaking, the general idea is to count a larger set of distinct objects and then use the Division Principle to "merge" the ones that are not significantly different from each other.

Theorem 2.20 (Division Principle)**.** If a finite set A is the union of n pairwise disjoint subsets each with d elements, then $|A| = n \cdot d$. Equivalently, $n = |A|/d$.

Figure 2.4 illustrates the Division Principle, where the set A contains 12 elements that are partitioned into $n = 3$ pairwise disjoint subsets, each of size $d = 4$. In this case, we see that $|A|/d = 12/4 = 3 = n$.

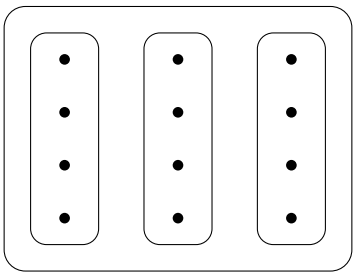


Figure 2.4: Visualization of the Division Principle with $|A| = 12$, $n = 3$, and $d = 4$.

Problem 2.21. How many ways can the letters of the word FLAGSTAFF be arranged?

Problem 2.22. Six friends sit around a circlular table to play a game. Rotations of the group do not constitute different seating orders.

(a) How many circular seating arrangements are there?

(b) How many circular seating arrangements are there if Sally and Maria always sit next to each other?

For $n \in \mathbb{N} \cup \{0\}$, a **set partition** of $[n]$ is a collection of nonempty pairwise disjoint subsets of $[n]$ whose union is $[n]$. Each subset in a set partition is called a **block**. For example, there are 7 set partitions of $[4]$ with 2 blocks, namely:

$$\begin{array}{ll} \{\{1,2,3\},\{4\}\} & \{\{1,2,4\},\{3\}\} \\ \{\{1,3,4\},\{2\}\} & \{\{2,3,4\},\{1\}\} \\ \{\{1,2\},\{3,4\}\} & \{\{1,3\},\{2,4\}\} \\ \{\{1,4\},\{2,3\}\} & \end{array}$$

The empty set has exactly one set partition, namely the empty collection of blocks, $\varnothing$. This partition has zero blocks; it is not $\{\varnothing\}$, since blocks must be nonempty.

We define the **Stirling numbers** (of the second kind) via

$$\left\{ {n \atop k} \right\} := \text{number of set partitions of } [n] \text{ with } k \text{ blocks.}$$

I usually pronounce this as "n Stirling k". Based on the information above, we know $\left\{ {4 \atop 2} \right\} = 7$. For $n \in \mathbb{N} \cup \{0\}$, we define the nth **Bell number** B_n to be the total number of set partitions of $[n]$ into any number of blocks. In particular, $B_0 = 1$, since the empty set has exactly one set partition.

The next problem is quite involved but provides an opportunity to put all of the counting principles to use.

Problem 2.23. Answer each of the following.

(a) Explain why $\left\{ {n \atop 1} \right\} = 1 = \left\{ {n \atop n} \right\}$ for every $n \in \mathbb{N}$.

(b) Explain why $\left\{ {n \atop 2} \right\} = 2^{n-1} - 1$ for every $n \in \mathbb{N}$.

(c) Explain why the Stirling numbers satisfy the following recurrence for $1 \leq k \leq n$:

$$\left\{ {n \atop k} \right\} = \left\{ {n-1 \atop k-1} \right\} + k \left\{ {n-1 \atop k} \right\}.$$

(d) Compute the Bell number B_4 by hand by writing down all of the appropriate set partitions.

(e) For $n \geq 1$, explain why

$$B_n = \sum_{k=1}^{n} \left\{ {n \atop k} \right\}.$$

Problem 2.24. Notice that back in Problem 2.11 we avoided counting surjective functions. We now have the tools to at least describe the count. Let $n, k \in \mathbb{N}$ wih $k \leq n$. How many surjective functions are there from $[n]$ to $[k]$?

Problem 2.25. Imagine we have $n \in \mathbb{N}$ people competing in a contest where ties are allowed.

(a) Explain why the number of final rankings is given by

$$\sum_{k=1}^{n} k! \left\{ {n \atop k} \right\}.$$

(b) Now, imagine that the people are numbered 1 through n. How many final rankings are possible under the proviso that person i's ranking is less than or equal to person $(i+1)$'s ranking? Such rankings are called **weakly increasing**.

2.3 Bijections

The next theorem states an important counting technique, which we refer to as the **Bijection Principle**.

Theorem 2.26 (Bijection Principle)**.** If there exists a bijection between finite sets A and B, then $|A| = |B|$.

For $n \in \mathbb{N} \cup \{0\}$, a **lattice path** of length n is a path in the integer lattice that begins at the origin and consists of n steps, each of which moves one unit north or one unit east. A lattice path of length n with r north steps is said to have rank r. Figure 2.5 provides an example of a lattice path of length 5 with rank 2.

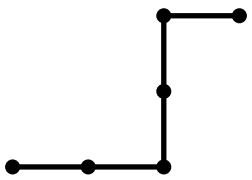


Figure 2.5: A lattice path of length 5 with rank 2.

For $n \in \mathbb{N}$ and $0 \leq r \leq n-1$, a **sticks and stones configuration** with n stones and r sticks is a string of n $\circ$'s and r |'s such that every | is framed by two $\circ$'s. For example, $\circ \mid \circ\circ\circ \mid \circ\circ$ is a sticks and stones configuration with 6 stones and 2 sticks.

For $n \in \mathbb{N} \cup \{0\}$, a **sparse strip** of length n is a $1 \times n$ grid such that each square in the grid is colored white or gray with the proviso that two gray squares are never adjacent. A sparse strip with r gray squares is said to have rank r. Figure 2.6 provides an example of a sparse strip of length 5 with rank 2.



Figure 2.6: A sparse strip of length 5 with rank 2.

Problem 2.27. Describe bijections between the following pairs of collections.

(a) For $n \in \mathbb{N} \cup \{0\}$, the collection of subsets of $[n]$ and the collection of binary strings of length n.

(b) For $n \in \mathbb{N} \cup \{0\}$, the collection of subsets of $[n]$ and the collection of lattice paths of length n.

(c) For $n \in \mathbb{N}$, the collection of compositions of n and the collection of sticks and stones configurations with n stones.

(d) For $n \in \mathbb{N}$, the collection of pebbles piles with n pebbles and the collection of sticks and stones configurations with n stones.

(e) For $n \in \mathbb{N} \cup \{0\}$, the collection of pebble pile configurations with $n+1$ pebbles and the collection of subsets of $[n]$.

(f) For $n \in \mathbb{N}$, the collection of nonattacking rook arrangements on an $n \times n$ board and the collection of bijections from $[n]$ to $[n]$.

(g) For $n \in \mathbb{N}$, the collection of outcomes of a contest with n people with ties allowed (see part (a) of Problem 2.25) and the collection of surjective functions with domain $[n]$.

(h) For $n \in \mathbb{N}$, the collection of outcomes of a contest with n people as described in part (b) of Problem 2.25 and the collection of sticks and stones configurations with n stones.

(i) For $n \in \mathbb{N}$, the collection of compositions of n with parts of size 1 and 2 only (see Problem 2.14) and the collection of domino tilings of a $2 \times n$ grid (see Problem 2.16).

(j) For $n \in \mathbb{N}$, the collection of domino tilings of a $2 \times n$ grid and the collection of sparse strips of length $n-1$.

In part (a) of Problem 2.27 you likely described a bijection between the collection of subsets of $[n]$ and the collection of binary strings of length n that also respected some additional structure. In particular, your bijection probably mapped k-subsets to binary strings with k ones. This is an example of a "nice" bijection, i.e., a set "isomorphism" that preserves combinatorial structure. In combinatorics, we really like these kinds of bijections.

Remark. Throughout these notes, we sometimes identify combinatorial objects with simpler representations when a specified bijection allows us to recover either uniquely from the other. Under this convention, we may say that an object "is" its representation, rather than repeatedly saying that it "corresponds uniquely to" that representation. For example, we may identify a nested ordered pair $(a,(b,c)) \in A \times (B \times C)$ with the ordered triple $(a,b,c) \in A \times B \times C$. These are different formal objects, but each uniquely determines the other. Such identifications may suppress braces, commas, or other notation, but they must retain enough information to recover the original object.

Problem 2.28. Using the Bijection Principle together with the previous problem and problems that appeared earlier, enumerate the following collections.

(a) For $n \in \mathbb{N} \cup \{0\}$, lattice paths of length n.

(b) For $n \in \mathbb{N}$, sticks and stones configurations with n stones.

(c) For $n \in \mathbb{N}$, the collection of surjective functions with domain $[n]$.

(d) For $n \in \mathbb{N} \cup \{0\}$, sparse strips of length n.

2.4 Combinations and Permutations

If A is a set and $B \subseteq A$ with $|B| = k$, we refer to B as a k-**subset** of A. The collection of all k-subsets of A is defined via

$$\binom{A}{k} := \{B \subseteq A \mid |B| = k\}.$$

For $n, k \in \mathbb{Z}$, the **binomial coefficients** are defined via

$$\binom{n}{k} := \begin{cases} \text{number of } k\text{-subsets of an } n\text{-element set}, & \text{if } 0 \leq k \leq n \\ 0, & \text{otherwise.} \end{cases}$$

In particular, if $|A| = n$, then $|\binom{A}{k}| = \binom{n}{k}$. We read "$\binom{n}{k}$" as "$n$ choose k". Alternate notations for binomial coefficients include $C(n,k)$ and ${}_nC_k$. We will see later why $\binom{n}{k}$ is referred to as a binomial coefficient.

Example 2.29. If $A = \{a, b, c, d\}$, then

$$\binom{A}{2} = \{\{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}, \{c,d\}\},$$

which implies that $\binom{4}{2} = 6$.

Problem 2.30. Let A be a finite set with $|A| = n \in \mathbb{N} \cup \{0\}$.

(a) What is $\binom{A}{0}$?

(b) What is $\binom{n}{0}$? Where do these values appear in Pascal's Triangle?

(c) What is $\binom{A}{n}$?

(d) What is $\binom{n}{n}$? Where do these values appear in Pascal's Triangle?

If we let n and k vary, we can organize the (nonzero) binomial coefficients in a triangular array, often referred to as **Pascal's Triangle**. See Table 2.2.

$n \backslash k$	0	1	2	3	4	5	6
0	1						
1	1	1					
2	1	2	1				
3	1	3	3	1			
4	1	4	6	4	1		
5	1	5	10	10	5	1	
6	1	6	15	20	15	6	1

Table 2.2: Pascal's Triangle of binomial coefficients.

Problem 2.31. Explain why the row sums in Pascal's Triangle are powers of two. In particular, for $n \geq 0$, explain why

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

We say that the binomial coefficients $\binom{n}{k}$ are a **refinement** of 2^n. This illustrates a general phenomenon we will witness over and over again.

Problem 2.32. Using the definition in terms of k-subsets (as opposed to some formula you might know involving factorials) together with a bijection, prove that for $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}.$$

This result is sometimes referred to as the **symmetry identity for binomial coefficients** (or simply **binomial symmetry**). Can you explain why it has this name?

The next result is called **Pascal's Recurrence** (or **Pascal's Identity**).

Problem 2.33 (Pascal's Recurrence)**.** Using the definition in terms of k-subsets, prove that for $n \in \mathbb{N}$ and $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

The next problem illustrates a powerful idea: If we count the same collection in two different ways, then the two answers must agree.

Problem 2.34. Explain why

$$1 + 2 + \cdots + n = \binom{n+1}{2}$$

by counting the same collection in two different ways.

For $n \in \mathbb{N}$, the number defined by $T_n := 1 + 2 + \cdots + n$ is called the nth **triangular number** (due to the shape we get by representing each number in the sum by a vertical stack of balls). We also set $T_0 = 0$ by convention.

Problem 2.35. For $n \in \mathbb{N}$ and $1 \leq k \leq n$, prove that the number of compositions of n with k parts is $\binom{n-1}{k-1}$.

For $k \in \mathbb{N}$ and a nonempty set A, a k-**permutation** of A is an injective function $w : [k] \to A$. The set of all k-permutations of A is denoted by $S_{A,k}$. If A happens to be the set $[n]$, we use the notation $S_{n,k}$. And if $n = k$, we write $S_n := S_{n,n}$ and refer to each n-permutation in S_n as a **permutation**. It's not coincidence that this notation coincides with the notation for the symmetric group that one encounters in abstract algebra. Define

$$P(n,k) := |S_{n,k}|.$$

By convention, we set $P(n,0) := 1$, including the case when $n = 0$.

Problem 2.36. For $1 \leq k \leq n$, describe a bijection between $S_{n,k}$ and the collection of nonattacking rook arrangments on a $n \times k$ board.

In light of Problems 2.12 and 2.36, we can conclude that

$$P(n,k) = n \cdot (n-1) \cdots (n+1-k) = \frac{n!}{(n-k)!}.$$

Problem 2.37. For $0 \leq k \leq n$, prove that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Problem 2.38. I'm in the mood for a cup of coffee! Let's say that the nearest place to get a cup of coffee is four blocks East and three blocks North. Assuming I only walk East or North, how many different routes can I take to get there?

Problem 2.39. How many binary strings of length 6 have an odd number of 0's?

Problem 2.40. Show that the Fibonacci numbers satisfy the following identity for $n \geq 0$:

$$f_n = \sum_{k \geq 0} \binom{n-k}{k}.$$

Hint: There are a few natural approaches. One elegant approach would be to utilize a combinatorial argument with one of the collections of compositions in Problem 2.14.

What is the previous result saying in terms of Pascal's Triangle?

Problem 2.41. Recall the definition of the Bell numbers (where $B_0 = 1$). Prove that for $n \geq 1$, we have

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k.$$

Hint: There is a block containing 1. Consider cases based on how many elements this block contains. Let k be the number of elements that are not in the block containing 1.

2.5 The Binomial Theorem

Recall that $\binom{n}{k}$ counts the number of subsets of size k taken from a set of size n. Why are these numbers called binomial coefficients? In general a binomial is just a polynomial with two terms. Let's explore a bit. We see that

$$\begin{aligned}(x+y)^2 &= (x+y)(x+y) \\ &= xx + xy + yx + yy \\ &= x^2 + 2xy + y^2.\end{aligned}$$

The coefficients in this expansion are $1, 2, 1$. Hey, we saw those in row $n = 2$ of Pascal's Triangle! Let's try a larger example. We see that

$$\begin{aligned}(x+y)^3 &= (x+y)^2(x+y)\\ &= (xx+xy+yx+yy)(x+y)\\ &= xxx+xyx+yxx+yyx+xxy+xyy+yxy+yyy\\ &= x^3+3x^2y+3xy^2+y^3.\end{aligned}$$

This time the coefficients are $1, 3, 3, 1$, which is the next row of Pascal's Triangle. What's going on here?

Consider the expansion of $(x+y)^n$. The key observation is that *before* commuting factors and collecting like terms, the terms in the expansion consist of *all possible* ordered products (i.e., words) where we choose either x or y from each factor. Each such term will consist of n letters. In particular, if there are k copies of x in a term, there will be $n-k$ copies of y. Moreover, every possible configuration of k copies of x (i.e., location of the x's in the term *before* doing any commuting with x and y) will be represented. This means there will be precisely $\binom{n}{k}$ many terms with k copies of x (and $n-k$ copies of y). Thus, when we commute and then collect like terms, the coefficient on $x^k y^{n-k}$ will indeed be $\binom{n}{k}$. This leads us to the following remarkable fact, known as the **Binomial Theorem**.

Theorem 2.42 (Binomial Theorem)**.** For $n \in \mathbb{N} \cup \{0\}$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Problem 2.43. Reprove that claim in Problem 2.31 using the Binomial Theorem.

It follows from the Binomial Theorem that

$$(1+t)^n = \sum_{k=0}^{n} \binom{n}{k} t^k = \binom{n}{0} t^0 + \binom{n}{1} t^1 + \cdots + \binom{n}{n} t^n.$$

That is, the expression $(1+t)^n$ encodes the entries in the nth row of Pascal's Triangle. In particular, $\binom{n}{k}$ is the coefficient on t^k in the expansion of $(1+t)^n$. This is our first example of what is called a **generating function**.

Chapter 3

Generating Functions

3.1 Ordinary Generating Functions

The (**ordinary**) **generating function** (or **generating series**) for a sequence $a_0, a_1, a_2, \ldots$ is the formal series

$$\sum_{k\geq 0} a_k t^k = a_0 + a_1 t^1 + a_2 t^2 + \cdots$$

In this context, we treat t as an indeterminate. Two formal power series are equal precisely when their corresponding coefficients are equal. Under most circumstances, we will not be concerned with issues related to convergence of the series. For a finite sequence, we take all subsequent terms to be zero.

A generating function is an algebraic device for encoding a sequence. Herbert Wilf wrote,

> *A generating function is a clothesline on which we hang up a sequence of numbers for display.*

We define operations on formal power series to agree with the usual operations on polynomials. Below, we define scalar multiplication, addition, and multiplication, respectively. In each case, including the product, each coefficient on the right-hand side involves only finitely many terms, so these operations require no assumptions about convergence.

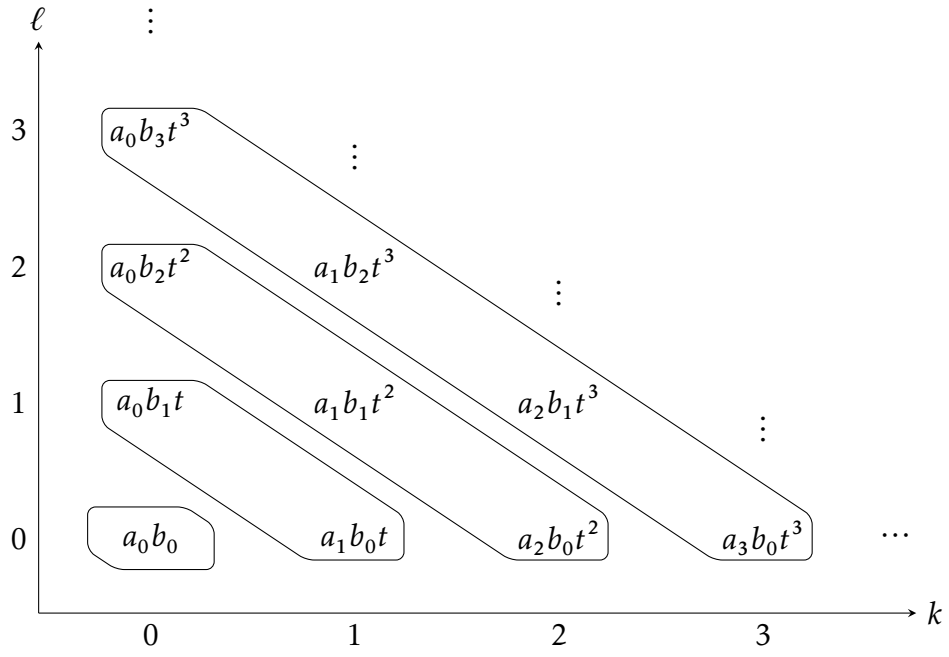
Definition 3.1. Let $a_0, a_1, a_2, \ldots$ and $b_0, b_1, b_2, \ldots$ be two sequences of numbers and let c be a constant. Then:

(a) $c \sum_{k\geq 0} a_k t^k = \sum_{k\geq 0} c a_k t^k$;

(b) $\left(\sum_{k\geq 0} a_k t^k\right) + \left(\sum_{k\geq 0} b_k t^k\right) = \sum_{k\geq 0} (a_k + b_k) t^k$;

(c) $\left(\sum_{k\geq 0} a_k t^k\right) \cdot \left(\sum_{\ell\geq 0} b_\ell t^\ell\right) = \sum_{j\geq 0} \left(\sum_{k+\ell=j} a_k b_\ell\right) t^j = \sum_{j\geq 0} \left(\sum_{k=0}^{j} a_k b_{j-k}\right) t^j$.

Figure 3.1 illustrates the multiplication in Definition 3.1(c): terms on the same diagonal have the same total degree $j = k + \ell$. Summing along that diagonal gives the coefficient $\sum_{k=0}^{j} a_k b_{j-k}$ of t^j.



$$a_0b_0 + (a_0b_1 + a_1b_0)t + (a_0b_2 + a_1b_1 + a_2b_0)t^2 + \cdots$$

Figure 3.1: Multiplying formal power series by collecting like terms. The entry at (k, ℓ) is $a_k b_\ell t^{k+\ell}$; the outlined groups collect the terms of degrees 0, 1, 2, and 3.

Example 3.2. Here are two examples of generating functions.

(a) The **geometric series** satisfies

$$1 + t + t^2 + \cdots = \sum_{k \geq 0} t^k = \frac{1}{1-t}.$$

Each expression above encodes the sequence $1, 1, 1, \ldots$. The expression $\frac{1}{1-t}$ on the far right is a **closed form** for the generating function. In the context of calculus, you may remember that the equality above only holds for $|t| < 1$. However, in our formal setting, each expression in the equation above should be interpreted as the multiplicative inverse of $1 - t$. Indeed, we have

$$\begin{aligned}(1-t)(1+t+t^2+\cdots) &= (1+t+t^2+\cdots) - t(1+t+t^2+\cdots) \\ &= (1+t+t^2+\cdots) - (t+t^2+\cdots) \\ &= 1.\end{aligned}$$

(b) For $n \in \mathbb{N} \cup \{0\}$, the Binomial Theorem tells us that

$$\binom{n}{0}t^0 + \binom{n}{1}t^1 + \cdots + \binom{n}{n}t^n = \sum_{k=0}^{n}\binom{n}{k}t^k = (1+t)^n.$$

For a fixed n, this is the generating function for the finite sequence $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$. The expression $(1+t)^n$ on the far right is a closed form for the generating function.

Problem 3.3. What sequence is $F(t) = \dfrac{1}{1+t}$ the generating function for?

Problem 3.4. What sequence is $F(t) = \dfrac{1}{t-3}$ the generating function for?

Problem 3.5. What sequence is $F(t) = \dfrac{1}{(1-t)^2}$ the generating function for?

Problem 3.6. Find a closed form for the generating function for the sequence $0, 1, 2, 3, \ldots$.

The next problem tells us that given the generating series for a sequence of numbers, we can obtain the generating series for the sequence of partial sums by simply multiplying by the geometric series.

Problem 3.7. Prove that if $F(t) = \sum_{k\geq 0} a_k t^k$, then

$$\sum_{k\geq 0}(a_0 + a_1 + \cdots + a_k)t^k = \frac{F(t)}{1-t}.$$

Problem 3.8. Explain why Problem 3.5 is a special case of Problem 3.7.

Recall that for $n \in \mathbb{N}$, the nth **triangular number** is defined via

$$T_n := 1 + 2 + \cdots + n$$

and we also set $T_0 = 0$ by convention.

Problem 3.9. Find a closed form for the generating function for the triangular numbers such that T_n is the coefficient on t^n.

Problem 3.10. What sequence is

$$F(t) = \frac{1}{1 - 5t + 6t^2}$$

the generating function for?

Hint: One approach is to use a partial fraction decomposition. Alternatively, you could use the formula for the product of two sums together with the following fact that you may freely use:

$$(x-y)(x^k + x^{k-1}y + x^{k-2}y^2 + \cdots + xy^{k-1} + y^k) = x^{k+1} - y^{k+1}.$$

We define differentiation and integration of generating series coefficientwise, without requiring convergence. For these operations, we assume that the coefficients are real numbers.

Definition 3.11. If $a_0, a_1, a_2, \ldots$ is a sequence of numbers, then we define the following.

(a) $\dfrac{d}{dt}\left[\sum_{k\geq 0} a_k t^k\right] := \sum_{k\geq 1} k a_k t^{k-1} = \sum_{k\geq 0} (k+1) a_{k+1} t^k$;

(b) $\displaystyle\int \sum_{k\geq 0} a_k t^k \, dt := C + \sum_{k\geq 0} \frac{a_k}{k+1} t^{k+1}$, where C is an arbitrary constant;

(c) $\displaystyle\int_0^t \sum_{k\geq 0} a_k x^k \, dx := \sum_{k\geq 0} \frac{a_k}{k+1} t^{k+1}$.

Differentiating any antiderivative of a generating series recovers the original series. In part (b), we obtain a family of generating series, each of which is called an **antiderivative**. In part (c), we obtain the unique antiderivative with constant term zero. Using the definitions above, we can obtain many of the familiar derivative and integral formulas from calculus. For example, if $A(t)$ and $B(t)$ are generating series, then

$$\frac{d}{dt}\big[A(t)B(t)\big] = A'(t)B(t) + A(t)B'(t).$$

The proof involves comparing coefficients of t^k on both sides.

To differentiate a closed form, we can use rules derived from formal differentiation. For example, if $F(t) = P(t)/Q(t)$, where P and Q are polynomials and $Q(0) \neq 0$, then Q has a formal multiplicative inverse. Differentiating $Q(t)F(t) = P(t)$ using the formal product rule yields the usual quotient rule:

$$F'(t) = \frac{P'(t)Q(t) - P(t)Q'(t)}{Q(t)^2}.$$

Problem 3.12. Find a closed form for the generating function for the sequence

$$0, 1, 4, 9, 16, \ldots.$$

Hint: Start with

$$A(t) = \sum_{k\geq 0} k t^k = \frac{t}{(1-t)^2},$$

the generating function for $0, 1, 2, 3, \ldots$. Differentiate $A(t)$, then determine how to shift the powers of t to obtain the desired sequence.

Given a closed form for a generating function, we can always recover the terms of the corresponding sequence by taking successive formal derivatives and extracting their constant terms. This gives a formal version of the familiar Maclaurin expansion from calculus. Here, the closed form is understood to represent a formal power series; evaluating such a series at $t = 0$ means taking its constant term.

Theorem 3.13 (Formal Maclaurin expansion)**.** Let $F(t) = \sum_{k\geq 0} a_k t^k$ be a formal power series with real coefficients. For every integer $k \geq 0$,

$$a_k = \frac{F^{(k)}(0)}{k!},$$

where $F^{(k)}$ denotes the kth formal derivative and $F^{(0)} = F$. Consequently,

$$F(t) = \sum_{k\geq 0} \frac{F^{(k)}(0)}{k!} t^k.$$

Proof. For each integer $k \geq 0$, repeated application of the definition of formal differentiation gives

$$F^{(k)}(t) = \sum_{j\geq k} \frac{j!}{(j-k)!} a_j t^{j-k}.$$

The constant term occurs when $j = k$ and is $k!a_k$. Thus $F^{(k)}(0) = k!a_k$, proving the coefficient formula and hence the series identity. □

Every formal power series can be differentiated arbitrarily many times by definition, so this theorem requires neither convergence nor Taylor's Theorem.

If we instead interpret the closed form as a function of a real variable, convergence becomes a separate issue. An analytic function near 0 equals its Maclaurin series in some neighborhood of 0; rational functions with denominator nonzero at 0 have this property. Infinite differentiability alone does not guarantee equality with the Maclaurin series: to obtain that equality from Taylor's Theorem, the Taylor remainders must tend to zero. The good news is that we do not really need to worry about these details.

In the next two problems, use the familiar derivatives from calculus to find the Maclaurin series, then regard these expansions as formal generating series.

Problem 3.14. Find the Maclaurin series for $F(t) = e^t$. What sequence is $F(t)$ the generating function for?

Problem 3.15. Find the Maclaurin series for $F(t) = \sin(t)$. What sequence is $F(t)$ the generating function for?

Problem 3.16. Use formal integration to find a closed form for the ordinary generating function for the sequence $1, \frac{1}{2}, \frac{1}{6}, \frac{1}{24}, \frac{1}{120}, \ldots$, whose kth term is $a_k = \frac{1}{(k+1)!}$ for $k \geq 0$.

Hint: Integrate the formal exponential series from 0 to t, then shift the powers of t so that the constant term is 1. Before shifting down by one power of t, check that the series has constant term zero. In this situation, division by t means lowering each exponent by one, rather than multiplying by a formal power-series inverse of t, which does not exist.

Problem 3.17. How many ways can you make exactly one dollar using pennies, nickels, dimes, and quarters? Assume that you have an unlimited supply of each denomination,

with values 1, 5, 10, and 25 cents, respectively. Coins of the same denomination are indistinguishable, and the order of the coins does not matter. For each integer $k \geq 0$, let a_k be the number of ways to make k cents using these denominations, and define

$$C(t) = \sum_{k \geq 0} a_k t^k.$$

(a) For a coin worth d cents, write a generating series that records the possible total values obtained using only that denomination. Find a closed form for this series.

(b) Use your answer to part (a) to express $C(t)$ as a product of four generating series, and then find a closed form. Explain why the coefficient of t^k counts the desired collections of coins.

(c) Using technology, find the coefficient a_{100} of t^{100} in the corresponding Maclaurin series. How many ways are there to make a dollar using pennies, nickels, dimes, and quarters? In WolframAlpha, you can use the following command:

```
SeriesCoefficient[C(t),{t,0,100}].
```

Let's close out the section with two problems. The first reviews earlier counting techniques, and the second develops a generating function for multisets.

Define a **multiset** M to be an unordered collection of elements that may be repeated. In this section, all multisets are finite. To distinguish between sets and multisets, we will use the notation {} versus {{}}. For example, $M = \{\{a, a, a, b, c, c\}\}$ is a multiset (not to be confused with the set containing the set $\{a, b, c\}$; mathematical notation is both awesome and frustrating at times). The primes occurring in the prime factorization of a natural number greater than 1 is an example of multiset. The cardinality of a multiset is its number of elements counted with multiplicity. So, in our example, $|M| = 3 + 1 + 2 = 6$. If S is a set, then M is a **multiset on** S if every element of M is an element of S. For integers $n, k \geq 0$, with $[0] = \varnothing$, define

$$\left(\!\binom{n}{k}\!\right) := \text{number of multisets on } [n] \text{ of size } k.$$

The only multiset on the empty set is the empty multiset, which has size zero. Thus,

$$\left(\!\binom{0}{0}\!\right) = 1 \qquad \text{and} \qquad \left(\!\binom{0}{k}\!\right) = 0 \quad \text{for } k \geq 1.$$

Problem 3.18. For integers $n \geq 1$ and $k \geq 0$, prove that

$$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}.$$

Hint: Here is one possible approach. Construct a bijection from the collection of multisets on $[n]$ of size k to the collection of subsets of $[n + k - 1]$ of size k. To do this, assume your multisets on $[n]$ are canonically written in nondecreasing order, say $\{\{m_1, m_2, \ldots, m_k\}\}$, where $m_1 \leq m_2 \leq \cdots \leq m_k$. Try to map each such multiset to a k-subset of $[n + k - 1]$.

Problem 3.19. For $n \in \mathbb{N} \cup \{0\}$, let $M_n(t)$ denote the generating function for the number of multisets of $[n]$ of size k:

$$M_n(t) := \sum_{k \geq 0} \left(\!\binom{n}{k}\!\right) t^k.$$

Also, define $\mathcal{M}_n := \{A \mid A \text{ is a multiset on } [n]\}$.

(a) Explain why $M_n(t) = \displaystyle\sum_{A \in \mathcal{M}_n} t^{|A|}$, where $|A|$ is the size of the multiset A.

(b) Explain why $\displaystyle\sum_{A \in \mathcal{M}_n} t^{|A|} = (1 + t + t^2 + \cdots)^n$.

(c) Explain why $M_n(t) = \dfrac{1}{(1-t)^n}$.

Hint: For part (b), notice that if you were to expand $(1 + t + t^2 + \cdots)^n$, each resulting term (prior to collecting like terms) corresponds to choosing a term from each of the n factors and then multiplying them together. Each such product of choices corresponds to a unique multiset. In particular, think of the ith factor of $(1+t+t^2+\cdots)^n$ as corresponding to $i \in [n]$. Choosing t^j in the ith factor corresponds to having i occur with multiplicity j in a multiset. When $n = 0$, use the convention that a product with no factors is equal to 1.

3.2 Exponential Generating Functions and Species

The **exponential generating function** (or **exponential generating series**) for a sequence $a_0, a_1, a_2, \ldots$ is the formal series

$$\sum_{k \geq 0} a_k \frac{t^k}{k!} = a_0 \frac{t^0}{0!} + a_1 \frac{t^1}{1!} + a_2 \frac{t^2}{2!} + \cdots.$$

Problem 3.20. For each function below, determine the sequence for which it is the exponential generating function.

(a) $F(t) = \dfrac{1}{1-t}$

(b) $F(t) = e^t$

(c) $F(t) = \sin(t)$

Problem 3.21. Determine all sequences whose exponential generating function $F(t)$ satisfies $F'(t) = F(t)$. How does the answer change if we swap out "exponential" with "ordinary"?

Problem 3.22. Let $A(t) = \sum_{k\geq 0} a_k \frac{t^k}{k!}$ and $B(t) = \sum_{k\geq 0} b_k \frac{t^k}{k!}$. Prove that

$$A(t)B(t) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \right) \frac{t^n}{n!}.$$

Given a finite set U, let F be a rule for producing a finite collection of labeled combinatorial structures on U, called F-**structures**. Let $F[U]$ denote the collection of F-structures on the set U. We assume that the number of F-structures on U is uniquely determined by $|U|$. We refer to F as a **combinatorial species** (or just **species** for short). For $n \in \mathbb{N} \cup \{0\}$, we define $F[n] := F[[n]]$ so that we do not need to write two pairs of square brackets. Note that $F[0] = F[\varnothing]$ and for $n \geq 1$, $F[n] = F[\{1, \ldots, n\}]$.

Here is a helpful analogy. Think of a species F as a vending machine that eats a finite set U. We hold out a bag to collect all the F-structures the machine can make out of U. The contents of our bag form the set $F[U]$. What happens when we feed the machine the empty set? For some species, the machine returns a single empty structure. If we represent that structure by $\varnothing$, our bag equals $\{\varnothing\}$, where the outer set braces represent the bag and the inner $\varnothing$ represents the empty structure. In this case, the bag contains one object, even though that object is empty. For other species, the machine returns nothing at all, so our bag is empty: $\{\} = \varnothing$. An empty structure in the bag is different from an empty bag! These are two possibilities; a species could also have more than one structure on the empty set.

Note: The phrase "combinatorial structures" is sufficiently vague here, but for our purposes, we will rely on our intuition. Some examples of labeled combinatorial structures are sets, subsets, permutations, graphs, planar binary trees, partitions, etc.

For $n \in \mathbb{N} \cup \{0\}$, the exponential generating function for the number of F-structures on $[n]$ is given by

$$F(t) := \sum_{n\geq 0} |F[n]| \frac{t^n}{n!}.$$

We say that $F(t)$ is the exponential generating function for the species F.

Using "F" in multiple ways (e.g., species F, F-structures $F[U]$, and exponential generating function $F(t)$) is an example of "overloading the operator", but this is common practice across mathematics. I typically object to using $F(t)$ instead of F to refer to the name of a function. However, I let it slide here due to the potential confusion between a species and its exponential generating function.

There are some details I am leaving out regarding the formal definition of species that you are more than welcome to look up. I'm fairly certain that leaving out these details will not do any damage. Fingers crossed!

Problem 3.23. Let E denote the **species of sets**. An E-structure on a finite set U is simply the set U itself, so $E[U] = \{U\}$, including when $U = \varnothing$. Find a closed form for the

exponential generating function

$$E(t) = \sum_{n \geq 0} |E[n]| \frac{t^n}{n!}.$$

Problem 3.24. Let S denote the **species of permutations**. An S-structure on a finite set U is a bijection from U to itself. That is, $S[U] = \{f : U \to U \mid f \text{ is a bijection}\}$. Find a closed form for the exponential generating function

$$S(t) = \sum_{n \geq 0} |S[n]| \frac{t^n}{n!},$$

where $|S[0]| = 1$ (i.e., the empty permutation is the only permutation we can make out of the empty set).

Problem 3.25. Let E_+ denote the **species of nonempty sets**. An E_+-structure on a finite nonempty set U is simply the set U itself, and there are no E_+-structures on the empty set. Thus,

$$E_+[U] = \begin{cases} \{U\}, & U \neq \varnothing, \\ \varnothing, & U = \varnothing. \end{cases}$$

(a) Determine $|E_+[n]|$ for each $n \in \mathbb{N} \cup \{0\}$. Explain how $E_+[0]$ differs from $E[0]$.

(b) Find a closed form for the exponential generating function

$$E_+(t) = \sum_{n \geq 0} |E_+[n]| \frac{t^n}{n!}.$$

(c) How is $E_+(t)$ related to $E(t)$?

The E_+ vending machine returns nothing when we insert the empty set: $E_+[\varnothing] = \varnothing$, so our bag is empty. In contrast, the E machine returns the empty set itself, so our bag contains one structure: $E[\varnothing] = \{\varnothing\}$. The previous problem generalizes to an arbitrary species.

Problem 3.26. Let F be any species. We define the species F_+ by restricting F to nonempty label sets. That is, for any finite set U,

$$F_+[U] = \begin{cases} F[U], & U \neq \varnothing, \\ \varnothing, & U = \varnothing. \end{cases}$$

Thus, an F_+-structure on a nonempty set is an F-structure on that set, and there are no F_+-structures on the empty set. In the vending-machine analogy, the F_+ machine behaves just like the F machine on nonempty inputs, but returns nothing when we insert the empty set, regardless of what the F machine would return.

(a) Express $F_+(t)$ in terms of $F(t)$ and $|F[0]|$. Under what condition is $F_+(t) = F(t) - 1$?

(b) Describe the species S_+ and find a closed form for its exponential generating function.

Problem 3.27. Fix $k \in \mathbb{N} \cup \{0\}$, and let E_k denote the **species of sets of size** k. An E_k-structure on a finite set U is simply the set U itself when $|U| = k$, and there are no E_k-structures on U when $|U| \neq k$. Thus,

$$E_k[U] = \begin{cases} \{U\}, & |U| = k, \\ \varnothing, & |U| \neq k. \end{cases}$$

(a) Determine $|E_k[n]|$ for each $n \in \mathbb{N} \cup \{0\}$. Include the case $k = 0$.

(b) Find a closed form for the exponential generating function

$$E_k(t) = \sum_{n \geq 0} |E_k[n]| \frac{t^n}{n!}.$$

Problem 3.28. Let U be a finite set with $n = |U|$ elements. We identify a linear order of U with a bijection $f : [n] \to U$, which we may represent using one-line notation: $f(1) \cdots f(n)$. For example, 231 represents a linear order f on $[3]$ given by $1 \mapsto 2$, $2 \mapsto 3$, and $3 \mapsto 1$. Let L denote the **species of linear orders**, so an L-structure on U is a linear order of U. There is one linear order on the empty set, represented by the empty word.

(a) Determine $|L[n]|$ for each $n \in \mathbb{N} \cup \{0\}$.

(b) Find a closed form for the exponential generating function

$$L(t) = \sum_{n \geq 0} |L[n]| \frac{t^n}{n!}.$$

(c) How is $L(t)$ related to $S(t)$?

Let F and G be two species. We define their **sum**, denoted $F + G$, by

$$(F + G)[U] := F[U] \sqcup G[U],$$

where $\sqcup$ denotes **disjoint union**. Informally, an $(F+G)$-structure on U consists of a choice of one of the two species together with a structure of that species on all of U. We retain which species was chosen, so an object that belongs to both $F[U]$ and $G[U]$ contributes two distinct structures to $(F + G)[U]$.

Problem 3.29. Prove that

$$(F + G)(t) = F(t) + G(t),$$

where F and G are two species.

Problem 3.30. Let $k, n \in \mathbb{N} \cup \{0\}$.

(a) Describe $(E_0 + E_1 + \cdots + E_k)[n]$.

(b) Determine $|(E_0 + E_1 + \cdots + E_k)[n]|$.

(c) Describe $E_0 + E_1 + \cdots + E_k$.

(d) Determine $(E_0 + E_1 + \cdots + E_k)(t)$.

Let F and G be two species. We define their **product**, denoted $F \cdot G$, by

$$(F \cdot G)[U] := \bigsqcup_{A \subseteq U} F[A] \times G[U \setminus A],$$

where $\sqcup$ denotes "disjoint union". Informally, an $(F \cdot G)$-structure is an ordered pair formed by an F-structure and a G-structure over complementary disjoint sets. That is, we put an F-structure on a subset of U and then a G-structure on that subset's complement.

Problem 3.31. Prove that

$$(F \cdot G)(t) = F(t)G(t),$$

where F and G are two species.

Problem 3.32. A **bicolored set** is a set in which every element is colored one of two colors (say red or blue). For example, $\{a, b, c\}$ and $\{a, b, c\}$ are two different bicolored sets.

(a) Explain why every collection of bicolored sets with n elements is in bijection with the collection of $(E \cdot E)$-structures on $[n]$, where E is the species introduced in Problem 3.23. It follows that the corresponding exponential generating function is $(E \cdot E)(t) = E(t)E(t) = e^t e^t = e^{2t}$.

(b) To really drive the point home, use Problem 3.22 to compute $E(t)E(t)$.

Problem 3.33. Let P denote the **species of subsets**. That is, a P-structure on $[n]$ is a subset of $[n]$. Find a closed form for $P(t)$.

Let F and G be two species. Further, let's suppose that $G[\varnothing] = \varnothing$ (i.e., there are no G-structures on the empty set). Using F and G, we can construct new structures by "composing" F and G.

We define their **composition**, denoted $F \circ G$, as follows: An $(F \circ G)$-structure on a finite set U is a triplet $s = (\pi, \phi, \gamma)$, where

- π is a partition on U;
- $\gamma = (\gamma_B)_{B \in \pi} \in \prod_{B \in \pi} G[B]$, so γ assigns a G-structure to each block of π;
- ϕ is an F-structure on the set of blocks of π.

If any block admits no G-structures, then no such γ exists, and this partition contributes no $(F \circ G)$-structures (see Problem 2.2).

This looks more complicated than it is. An $(F \circ G)$-structure of U simply means we put G-structures on each of the blocks of a partition of U and then put an F-structure on the collection of blocks.

Thus, for any finite set U,

$$(F \circ G)[U] = \left\{ (\pi, \phi, \gamma) \;\middle|\; \begin{array}{l} \pi \text{ is a partition of } U, \\ \phi \in F[\pi], \\ \gamma = (\gamma_B)_{B \in \pi}, \quad \gamma_B \in G[B] \text{ for every } B \in \pi \end{array} \right\}.$$

By grouping the last two entries of each triple, we obtain a canonical bijection

$$(F \circ G)[U] \longleftrightarrow \bigsqcup_{\pi \text{ partition of } U} F[\pi] \times \prod_{B \in \pi} G[B].$$

Here the disjoint union retains the indexing partition π as a tag, so an element in the component indexed by π is represented by $(\pi, (\phi, \gamma))$. The bijection is given by

$$(\pi, \phi, \gamma) \longmapsto (\pi, (\phi, \gamma)).$$

Remark. We will often describe an $(F \circ G)$-structure using a simpler representation of the tuple (π, ϕ, γ). That is, we may identify the tuple with this representation when each uniquely determines the other. For species, we also require these identifications to respect relabeling: changing the labels before or after passing to the representation gives the same result.

Example. An $(L \circ E_2)$-structure on a finite set U corresponds uniquely to an ordered list of unordered pairs that partition U. Think of forming teams of two and then lining up the teams. The E_2-structures make each block a pair, and the L-structure puts the collection of pairs in a linear order.

For example, on $U = [6]$, take

$$\pi = \big\{\{1,4\}, \{2,6\}, \{3,5\}\big\}.$$

Define the linear order $\phi : [3] \to \pi$ by

$$\phi(1) = \{2,6\}, \qquad \phi(2) = \{1,4\}, \qquad \phi(3) = \{3,5\}.$$

Each block $B \in \pi$ has a unique E_2-structure, namely B itself. Thus, with $\gamma = (\gamma_B)_{B \in \pi}$ defined by $\gamma_B = B$, the formal tuple

$$s = (\pi, \phi, \gamma)$$

represents the ordered list

$$\big(\{2,6\}, \{1,4\}, \{3,5\}\big).$$

Changing the order of the pairs changes the structure. Writing $\{6,2\}$ instead of $\{2,6\}$ does not. There are

$$|(L \circ E_2)[6]| = \binom{6}{2}\binom{4}{2}\binom{2}{2} = 90$$

such structures on $[6]$: choose the first pair, then the second, then the third.

On $[5]$, the entire collection is

$$(L \circ E_2)[5] = \big\{(\pi, \phi, (\gamma_B)_{B \in \pi}) \mid \pi \text{ partitions } [5],\ \phi \in L[\pi],\ \gamma_B \in E_2[B] \text{ for every } B \in \pi\big\} = \varnothing,$$

since a set with five elements cannot be partitioned into pairs. There are no tuples to list.

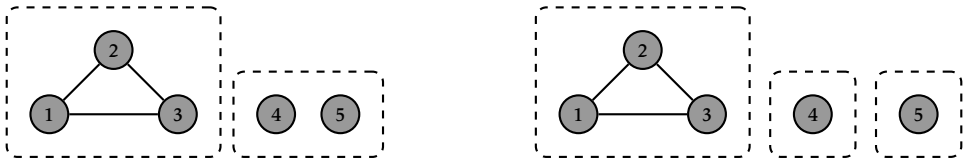
Remark. Why do we require $G[\varnothing] = \varnothing$? The blocks of a set partition are nonempty, so composition uses G-structures only on nonempty label sets. In contrast, the species product G^k (the product of k copies of G) allows empty parts. Requiring $G[\varnothing] = \varnothing$ ensures that these empty parts contribute no structures, allowing $G(t)^k$ to count ordered collections of k nonempty blocks carrying G-structures.

Example. Let $\mathcal{G}$ denote the **species of simple graphs**. A $\mathcal{G}$-structure on a finite set U is a graph with vertex set U whose edges are unordered pairs of distinct elements of U. There is one graph on the empty vertex set, namely the empty graph.

An $(E \circ \mathcal{G}_+)$-structure corresponds uniquely to a set partition together with a simple graph on each block. To satisfy the empty-set condition for composition, we use $\mathcal{G}_+$, the species of simple graphs with nonempty vertex sets. These graphs may have no edges. For example, on $[5]$, let

$$\pi = \{\{1, 2, 3\}, \{4, 5\}\}.$$

On the first block, construct the graph with edges $\{1, 2\}$, $\{2, 3\}$, and $\{1, 3\}$, and on the second block, construct the graph with no edges. In the formal tuple $s = (\pi, \phi, \gamma)$, γ records the graph on each block.



Two blocks Three blocks

Figure 3.2: Two distinct $(E \circ \mathcal{G}_+)$-structures on $[5]$ with the same underlying graph. Dashed boundaries indicate the blocks.

The left diagram in Figure 3.2 represents the structure just described. The right diagram represents a different structure, with partition

$$\pi' = \{\{1, 2, 3\}, \{4\}, \{5\}\}.$$

Both give the same graph on $[5]$ when we forget the blocks and take the union of the edges. Hence the graph alone does not determine the $(E \circ \mathcal{G}_+)$-structure: we must retain the partition as well.

Problem 3.34. Prove that

$$(F \circ G)(t) = F(G(t)),$$

where F and G are two species such that $G[\varnothing] = \varnothing$.

Example. Recall the earlier example in which $(L \circ E_2)$-structures correspond uniquely to ordered lists of unordered pairs. Since $E_2[\varnothing] = \varnothing$, Problem 3.34 applies. Using $L(t) = 1/(1-t)$ and $E_2(t) = t^2/2$, we obtain

$$(L \circ E_2)(t) = L(E_2(t)) = \frac{1}{1 - t^2/2}.$$

Expanding as a geometric series and then writing the result as an exponential generating series gives

$$\frac{1}{1-t^2/2} = \sum_{k\geq 0} \frac{t^{2k}}{2^k} = \sum_{k\geq 0} \frac{(2k)!}{2^k} \frac{t^{2k}}{(2k)!}.$$

Thus, there are $(2k)!/2^k$ structures on $[2k]$ and no structures on a label set of odd size. In particular, this recovers the earlier counts

$$|(L \circ E_2)[6]| = \frac{6!}{2^3} = 90 \qquad \text{and} \qquad |(L \circ E_2)[5]| = 0.$$

The constant term is 1, corresponding to the empty list of pairs on the empty set.

Problem 3.35. Let $\mathcal{C}$ denote the **species of cycles**. For a nonempty finite set U, a $\mathcal{C}$-structure on U is a cyclic ordering of all its elements, with orderings that differ by a cyclic rotation identified. For example,

$$\mathcal{C}[\{a,b,c\}] = \{(a,b,c),(a,c,b)\}.$$

These two cycles are depicted in Figure 3.3. There are no $\mathcal{C}$-structures on the empty set.

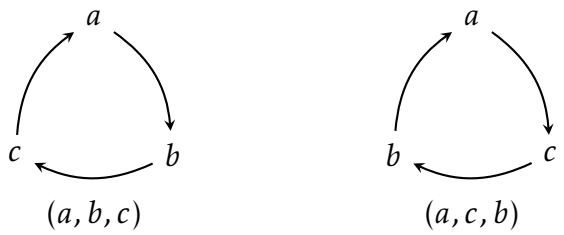


Figure 3.3: The two cycles on $\{a,b,c\}$. Arrows indicate the successor of each element.

Recall from abstract algebra that a permutation on a finite set U can be represented in cycle notation. For example, the permutation $w : [6] \to [6]$ with one-line notation $w = 345216$ can be written as the product of disjoint cycles

$$w = (1,3,5)(2,4)(6).$$

Here, $1 \mapsto 3 \mapsto 5 \mapsto 1$, $2 \mapsto 4 \mapsto 2$, and 6 is fixed. We include the singleton cycle (6) to account for every element of $[6]$.

(a) Explain why a permutation on a finite set U corresponds uniquely to an $(E \circ \mathcal{C})$-structure on U.

(b) Use part (a) and Problem 3.34 to find a closed form for $\mathcal{C}(t)$.

(c) Explain directly why $|\mathcal{C}[n]| = (n-1)!$ for $n \geq 1$.

(d) Expand your answer from part (b) as an exponential generating series and verify that it gives the same count as part (c).

Hint: Compute both sides of

$$\int_0^t \frac{1}{1-x}\,dx = \int_0^t \sum_{n\geq 0} x^n\,dx.$$

Problem 3.36. Recall the definition of set partition given earlier. Let Par denote the **species of set partitions**. Use Problem 3.34 to find a closed form for $\mathrm{Par}(t)$.

Problem 3.37. For $n \in \mathbb{N} \cup \{0\}$, an **ordered set partition** (or **ballot**) of $[n]$ is a set partition of $[n]$ together with a linear order on its collection of blocks. For $n \geq 1$, we represent it by a list $(B_1, B_2, \ldots, B_k)$ containing each block exactly once, where $1 \leq k \leq n$. For $n = 0$, the collection of blocks is empty and has a unique linear order, so the unique ordered set partition is the empty list $()$. Let Bal denote the **species of ordered set partitions**. Find a closed form for $\mathrm{Bal}(t)$.

Problem 3.38. Let $n \in \mathbb{N} \cup \{0\}$. For $n \geq 1$, a **Cayley permutation** on $[n]$ is a function $w : [n] \to [n]$ whose image is $[k]$ for some $1 \leq k \leq n$. We write w in one-line notation as $w(1) \cdots w(n)$. For $n = 0$, we take the empty word to be the unique Cayley permutation on $[0]$. Thus, a Cayley permutation is a word of positive integers, possibly empty, such that whenever b appears, every positive integer $a < b$ also appears.

Let Cay denote the **species of Cayley permutations**. Then $\mathrm{Cay}[n]$ is the collection of Cayley permutations on $[n]$. For example,

$$\mathrm{Cay}[3] = \{111, 112, 121, 122, 123, 132, 211, 212, 213, 221, 231, 312, 321\}.$$

(a) Construct a bijection between $\mathrm{Cay}[n]$ and $\mathrm{Bal}[n]$, the collection of ordered set partitions of $[n]$.

(b) Use this bijection and Problem 3.37 to find a closed form for $\mathrm{Cay}(t)$.

Let F be a species. We define its **derivative species**, denoted F', by

$$F'[U] = F[U \sqcup \{*\}],$$

where $*$ is one additional label not in the finite set U. Thus, an F'-structure on U is an F-structure on the enlarged label set $U \sqcup \{*\}$. The extra label is not counted among the labels of U and remains fixed when we relabel U. In particular, $F'[\varnothing] = F[\{*\}]$.

Problem 3.39. Let F be a species. Prove that

$$F'(t) = \frac{d}{dt}F(t),$$

where $F'(t)$ denotes the exponential generating function of the species F'.

Problem 3.40. Recall that L denotes the species of linear orders. Prove that

$$|(L \cdot L)[U]| = |L'[U]|$$

by describing a bijection between $(L \cdot L)[U]$ and $L'[U]$ for every finite set U, including $U = \varnothing$.

Problem 3.41. By Problems 3.39 and 3.40 and the product formula for species, we have $L'(t) = L(t)^2$. Independently verify this identity by differentiating the closed form for $L(t)$ and comparing the result with its square.

Let F be a species. The **pointed species** associated with F, denoted $F^\bullet$, is defined by

$$F^\bullet[U] = F[U] \times U$$

for each finite set U. An $F^\bullet$-structure is a pair (s, u) consisting of a structure $s \in F[U]$ and a chosen label $u \in U$, called the **distinguished element**. We think of u as the element we are pointing at. Unlike differentiation, pointing selects an existing label rather than adjoining an extra one. In particular, $F^\bullet[\varnothing] = \varnothing$, since there is no label to select.

Example 3.42. There are $3 \cdot 3! = 18$ pointed linear orders on $[3]$. Writing each linear order in one-line notation, we have

$$L^\bullet[3] = \begin{Bmatrix} (123,1), & (123,2), & (123,3), \\ (132,1), & (132,2), & (132,3), \\ (213,1), & (213,2), & (213,3), \\ (231,1), & (231,2), & (231,3), \\ (312,1), & (312,2), & (312,3), \\ (321,1), & (321,2), & (321,3) \end{Bmatrix}.$$

We can represent each pair (w, u) by placing a dot above the designated element u in the word w. Thus, we can simply describe this collection as the set of linear orders on $[3]$ with one designated element, indicated by a dot. In the same order as above, these are

$$\begin{Bmatrix} \dot{1}23, & 1\dot{2}3, & 12\dot{3}, \\ \dot{1}32, & 1\dot{3}2, & 13\dot{2}, \\ \dot{2}13, & 2\dot{1}3, & 21\dot{3}, \\ \dot{2}31, & 2\dot{3}1, & 23\dot{1}, \\ \dot{3}12, & 3\dot{1}2, & 31\dot{2}, \\ \dot{3}21, & 3\dot{2}1, & 32\dot{1} \end{Bmatrix}.$$

Problem 3.43. Recall that $\mathcal{C}$ denotes the species of cycles and L_+ denotes the species of linear orders on nonempty sets.

(a) For every finite set U, describe a bijection between $\mathcal{C}^\bullet[U]$ and $L_+[U]$.

(b) Use your bijection to find a closed form for $\mathcal{C}^\bullet(t)$.

Problem 3.44. Let F be a species. Explain why, for every $n \in \mathbb{N} \cup \{0\}$,

$$|F^\bullet[n]| = n|F[n]|.$$

Problem 3.45. Let F be a species, and recall that E_1 denotes the species of singleton sets. Prove that

$$|F^\bullet[U]| = |(E_1 \cdot F')[U]|$$

by describing a bijection between $F^\bullet[U]$ and $(E_1 \cdot F')[U]$ for every finite set U.

Problem 3.46. Recall that Par denotes the species of set partitions.

(a) For every finite set U, describe a bijection between $\mathrm{Par}^\bullet[U]$ and $(E_1 \cdot E \cdot \mathrm{Par})[U]$.

(b) Use your bijection to find a closed form for $\mathrm{Par}^\bullet(t)$.

Problem 3.47. Let $H = L \circ E_2$, so an H-structure corresponds uniquely to an ordered list of unordered pairs that partition the label set.

(a) For every finite set U, describe a bijection between $H^\bullet[U]$ and $(H \cdot E_2^\bullet \cdot H)[U]$.

(b) Determine $E_2^\bullet(t)$, and use part (a) to find a closed form for $H^\bullet(t)$.

Problem 3.48. Consider the species $\mathrm{Par}^\bullet$ and $E^\bullet \circ E_+$.

(a) Describe the structures of each species on a finite set U. In particular, explain what is distinguished in each case.

(b) For each set partition of $[3]$, determine how many structures of each species have that underlying partition. Use your counts to determine $|\mathrm{Par}^\bullet[3]|$ and $|(E^\bullet \circ E_+)[3]|$.

Chapter 4

Permutation Statistics

4.1 Eulerian Numbers

Recall that for a finite set U with $|U| = n$, the collection $S[U]$ consists of all bijections from U to itself, while $L[U]$ consists of all bijections from $[n]$ to U. When $U = [n]$, both collections consist of the same bijections:

$$S[n] = \{w : [n] \to [n] \mid w \text{ is a bijection}\} = L[n].$$

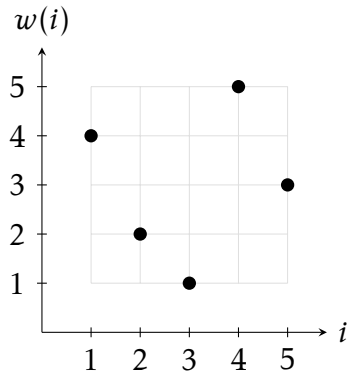
Thus, for $n \geq 1$, we can regard every permutation $w \in S[n]$ as a linear order and write it in **one-line notation** as $w = w(1)w(2)\cdots w(n)$. Conversely, a list containing each element of $[n]$ exactly once determines a unique permutation: the entry in position i is $w(i)$. This also holds for $n = 0$, where the unique bijection is represented by the empty word.

There are several other ways to represent a permutation $w \in S[n]$. In **cycle notation**, we record the disjoint cycles obtained by repeatedly applying w; each entry in a cycle maps to the next, and the last maps back to the first. The **graph of a permutation** consists of the points $(i, w(i))$ for $i \in [n]$, with inputs on the horizontal axis and outputs on the vertical axis. Since each input and each output occurs exactly once, there is exactly one point in each row and each column. These graphs are clearly in bijection with nonattacking rook arrangements having n rooks on an $n \times n$ board. A **permutation diagram** consists of two rows of n dots, each labeled $1, 2, \ldots, n$ from left to right, with an edge connecting i in the top row to j in the bottom row precisely when $w(i) = j$. We typically allow the edges to bend so that no three or more edges intersect at the same point.

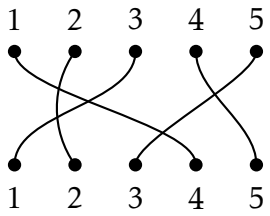
Example 4.1. Consider the permutation $w \in S[5]$ with one-line notation $w = 42153$. Then $w(1) = 4$, $w(2) = 2$, $w(3) = 1$, $w(4) = 5$, and $w(5) = 3$. In cycle notation, we have

$$w = (1, 4, 5, 3)(2).$$

This permutation has one cycle of length 4 and the fixed point 2. Its graph and permutation diagram appear in Figure 4.1. Notice that edges belonging to different cycles can cross.



(a) Graph of w.



(b) Permutation diagram of w.

Figure 4.1: Two representations of the permutation $w = 42153 = (1, 4, 5, 3)(2)$ from Example 4.1.

For any $w \in S[n]$, a **descent** is a position i with $1 \leq i \leq n-1$ such that $w(i) > w(i+1)$. In the graph of w, a descent occurs when the point in column i is higher than the point in the next column: moving from $(i, w(i))$ to $(i+1, w(i+1))$ goes downward. In the permutation diagram, a descent occurs when the edges from adjacent top vertices i and $i+1$ have their bottom endpoints in the reverse order. That is, if we assume the permutation diagram is drawn so that edges have no unnecessary crossings, we have a descent at i if the edge leaving i in the top row crosses the edge leaving $i+1$ in the top row.

The collection of descents for a permutation w is denoted by

$$\mathrm{Des}(w) := \begin{cases} \varnothing, & \text{if } n = 0, \\ \{i \in [n-1] \mid w(i) > w(i+1)\}, & \text{if } n \geq 1. \end{cases}$$

and the number of descents is

$$\mathrm{des}(w) := |\mathrm{Des}(w)|.$$

Example 4.2. The permutation $w = 42153$ from Example 4.1 (see Figure 4.1) has descents at positions 1, 2, and 4, since $4 > 2$, $2 > 1$, and $5 > 3$. Then $\mathrm{Des}(w) = \{1, 2, 4\}$ and $\mathrm{des}(w) = 3$.

Given $w = w(1) \cdots w(n) \in S[n]$, we define the **reverse** of w via

$$\mathrm{rev}(w) := w(n) \cdots w(1).$$

Problem 4.3. Let $w = 83572146 \in S[8]$.

(a) Find the cycle notation, graph, and permutation diagram for w.

(b) Find $\mathrm{Des}(w)$ and $\mathrm{des}(w)$.

(c) Find w^{-1}.

(d) Find the cycle notation, graph, and permutation diagram for w^{-1}.

(e) Find $\mathrm{Des}(w^{-1})$ and $\mathrm{des}(w^{-1})$.

(f) Find $\mathrm{rev}(w)$.

(g) Find the cycle notation, graph, and permutation diagram for $\mathrm{rev}(w)$.

(h) Find $\mathrm{Des}(\mathrm{rev}(w))$ and $\mathrm{des}(\mathrm{rev}(w))$.

Problem 4.4. Find all permutations in $S[n]$ with 0 descents.

Problem 4.5. What is the maximum number of descents attainable by a permutation in $S[n]$? How many permutations attain this maximum?

Problem 4.6. Let $w \in S[n]$.

(a) Is $\mathrm{Des}(w) = \mathrm{Des}(w^{-1})$? How about $\mathrm{des}(w) = \mathrm{des}(w^{-1})$? Justify your answer.

(b) What is the relationship between the graph of w and the graph of w^{-1}?

(c) What is the relationship between the permutation diagram of w and the permutation diagram of w^{-1}?

(d) What is the relationship between $\mathrm{Des}(w)$ and $\mathrm{Des}(\mathrm{rev}(w))$? How about $\mathrm{des}(w)$ and $\mathrm{des}(\mathrm{rev}(w))$?

(e) What is the relationship between the graph of w and the graph of $\mathrm{rev}(w)$?

(f) What is the relationship between the permutation diagram of w and the permutation diagram of $\mathrm{rev}(w)$?

For $n, k \in \mathbb{N} \cup \{0\}$, we define the **Eulerian numbers** via

$$\left\langle {n \atop k} \right\rangle := |\{w \in S[n] \mid \mathrm{des}(w) = k\}|.$$

That is, $\left\langle {n \atop k} \right\rangle$ counts the number of permutations in $S[n]$ with k descents.

Example 4.7. Let's compute some Eulerian numbers by brute force. For $n = 0$, the set $S[0]$ contains only the empty permutation, which has no descents. For $1 \leq n \leq 4$, Table 4.1 organizes the permutations in $S[n]$ by their number of descents, with one subtable for each value of n. Counting the entries in each column gives the corresponding Eulerian numbers, which are listed in Table 4.2. That table also includes the values for $n = 5$.

Problem 4.8. For integers $n \geq 1$ and $0 \leq k \leq n-1$, explain each of the following.

(a) $\left\langle {n \atop 0} \right\rangle = 1$.

(b) $\left\langle {n \atop n-1} \right\rangle = 1$.

(c) $\left\langle {n \atop k} \right\rangle = \left\langle {n \atop n-k-1} \right\rangle$.

What does part (c) say about each row in the array of Eulerian numbers?

$\mathrm{des}(w) = 0$
1

(a) Permutations in $S[1]$.

$\mathrm{des}(w) = 0$	$\mathrm{des}(w) = 1$
12	21

(b) Permutations in $S[2]$.

$\mathrm{des}(w) = 0$	$\mathrm{des}(w) = 1$	$\mathrm{des}(w) = 2$
123	132	321
	213	
	231	
	312	

(c) Permutations in $S[3]$.

$\mathrm{des}(w) = 0$	$\mathrm{des}(w) = 1$	$\mathrm{des}(w) = 2$	$\mathrm{des}(w) = 3$
1234	1243	1432	4321
	1324	2143	
	1342	2431	
	1423	3142	
	2134	3214	
	2314	3241	
	2341	3421	
	2413	4132	
	3124	4213	
	3412	4231	
	4123	4312	

(d) Permutations in $S[4]$.

Table 4.1: Permutations in $S[n]$ organized by number of descents for $1 \leq n \leq 4$.

$n\backslash k$	0	1	2	3	4
0	1				
1	1				
2	1	1			
3	1	4	1		
4	1	11	11	1	
5	1	26	66	26	1

Table 4.2: Eulerian numbers for $0 \leq n \leq 5$. Blank entries represent zeros.

Problem 4.9. For $n \geq 1$, what is $\sum_{k=0}^{n-1} \left\langle {n \atop k} \right\rangle$ equal to?

Problem 4.10. For $n \in \mathbb{N} \cup \{0\}$, prove that $\left\langle {n \atop 1} \right\rangle = 2^n - n - 1$. You may need to handle the cases for $n = 0$ and $n = 1$ separately.

The function

$$\mathrm{des} : \bigcup_{n \in \mathbb{N} \cup \{0\}} S[n] \to \mathbb{N} \cup \{0\}$$

is an example of a permutation statistic. In these notes, a **permutation statistic** is a function σ that assigns a nonnegative integer to each permutation with the additional restriction that σ assigns the value 0 to the empty permutation.

For a statistic σ, define

$$d_\sigma(n, k) := |\{w \in S[n] \mid \sigma(w) = k\}|.$$

Since $S[0]$ consists of the empty permutation and every permutation statistic assigns it the value 0, we have $d_\sigma(0, 0) = \left\langle {0 \atop 0} \right\rangle = 1$. More generally, $d_\sigma(0, k) = \left\langle {0 \atop k} \right\rangle$ for every $k \geq 0$.

When we organize the permutations in $S[n]$ according to σ, we obtain the counts $d_\sigma(n,k)$. For a fixed n, we say that the sequence determined by $d_\sigma(n,k)$ is the **distribution** of σ on $S[n]$.

For example, if we take σ to be des, then

$$d_\sigma(n,k) = \left\langle {n \atop k} \right\rangle.$$

And if we organize the permutations in $S[4]$ according to des, we get Table 4.1d, which yields

$$\left\langle {4 \atop 0} \right\rangle = 1, \left\langle {4 \atop 1} \right\rangle = 11, \left\langle {4 \atop 2} \right\rangle = 11, \left\langle {4 \atop 3} \right\rangle = 1.$$

The counts $d_\sigma(n,k)$ refine the total count $n!$ by recording how many permutations have each value of σ. Since every permutation has exactly one value of σ, we have

$$\sum_{k \geq 0} d_\sigma(n,k) = n!$$

for every $n \geq 0$. Only finitely many terms in this sum are nonzero.

For $n, k \in \mathbb{N} \cup \{0\}$, the numbers $d_\sigma(n,k)$ determine an array of values. Since $S[n]$ is finite and nonempty, each statistic σ has a minimum value m_n and a maximum value M_n on $S[n]$. Thus, $d_\sigma(n,k) = 0$ outside this range. In the examples we consider, every integer between m_n and M_n occurs as a value of σ. We typically restrict our attention to the portion of the array where $m_n \leq k \leq M_n$. For example, if we take σ to be des, then we obtain the array of Eulerian numbers as seen in Table 4.2. In this case, $m_0 = M_0 = 0$, while for $n \geq 1$ we have $m_n = 0$ and $M_n = n - 1$ (see Problems 4.4 and 4.5). Typically, we will encounter $m_n = 0$ or $m_n = 1$ while M_n will be something like $n-1, n, n+1$, or even $\binom{n}{2}$, depending on the specific permutation statistic.

We say that two permutation statistics σ and τ are **equidistributed** if

$$d_\sigma(n,k) = d_\tau(n,k)$$

for every $n, k \geq 0$. However, σ and τ need not agree on individual permutations.

In these notes, we call a permutation statistic σ **Eulerian** if there is a fixed nonnegative integer c such that

$$d_\sigma(n,k) = \left\langle {n \atop k-c} \right\rangle$$

for every $n \geq 1$ and $k \geq 0$. Here, c is independent of n and k, and Eulerian numbers with indices outside their valid range are understood to be zero. Typically, we have $c = 0$ or $c = 1$.

When $c = 0$, the definition of Eulerian says that σ and des are equidistributed. In particular, des itself is Eulerian with $c = 0$.

Problem 4.11. For any $w \in S[n]$, an **ascent** is a position i with $1 \leq i \leq n-1$ such that $w(i) < w(i+1)$. The collection of ascents for a permutation w is denoted by

$$\mathrm{Asc}(w) := \begin{cases} \varnothing, & \text{if } n = 0, \\ \{i \in [n-1] \mid w(i) < w(i+1)\}, & \text{if } n \geq 1. \end{cases}$$

and the number of ascents is

$$\mathrm{asc}(w) := |\mathrm{Asc}(w)|.$$

Prove that asc is Eulerian. Are des and asc equidistributed?

Problem 4.12. For any $w \in S[n]$, a **maximal increasing run** of w is a nonempty contiguous substring (of the one-line notation of w) whose entries are strictly increasing and which cannot be extended to the left or right while remaining strictly increasing. Note that maximal increasing runs can consist of a single value and the empty permutation has zero maximal increasing runs. The number of maximal increasing runs of a permutation w is denoted by $\mathrm{runs}(w)$. Prove that runs is Eulerian by showing that $d_{\mathrm{runs}}(n,k) = \left\langle {n \atop k-1} \right\rangle$ for $n \geq 1$. Compare the distributions in Table 4.3. Are des and runs equidistributed?

$d_{\mathrm{des}}(n,k) = \left\langle {n \atop k} \right\rangle$ for $n \geq 0$

$n\backslash k$	0	1	2	3	4	5
0	1					
1	1					
2	1	1				
3	1	4	1			
4	1	11	11	1		
5	1	26	66	26	1	

(a) Descents: $d_{\mathrm{des}}(n,k)$.

$d_{\mathrm{runs}}(n,k) = \left\langle {n \atop k-1} \right\rangle$ for $n \geq 1$

$n\backslash k$	0	1	2	3	4	5
0	1					
1		1				
2		1	1			
3		1	4	1		
4		1	11	11	1	
5		1	26	66	26	1

(b) Runs: $d_{\mathrm{runs}}(n,k)$.

Table 4.3: Distributions of descents and maximal increasing runs for $0 \leq n \leq 5$. Blank entries represent zeros.

Problem 4.13. For any $w \in S[n]$, an **excedance** is a position $i \in [n]$ such that $w(i) > i$. That is, the value in position i in one-line notation is larger than its position. The number of excedances of a permutation w is denoted $\mathrm{exc}(w)$. Prove that exc is Eulerian. Are des and exc equidistributed?

Hint: This one is tricky! Given a permutation w, write it in cycle notation, including 1-cycles. Moreover, write each cycle so that its least element appears first, and then write the collection of cycles such that the leading elements are in decreasing order. For example, if $w = 137482695$, then write w as $(5,8,9)(4)(2,3,7,6)(1)$. Now, define $f : S[n] \to S[n]$ where $f(w)$ is the permutation we obtain by writing w in the special cycle notation and then removing all parentheses and commas. For example, $f(137482695) = 589423761$. Explain why f is a bijection. The convention with the ordering of the cycles is crucial here. Then consider excedances of w versus ascents of $f(w)$. You should attempt to argue that an excedance in w yields an ascent in $f(w)$ and vice versa. Notice that ascents will only occur

within the cycles on the intermediate step of creating $f(w)$ due to our protocol for ordering the cycles. This approach is known as the "transformation fondamentale" (fundamental transformation) and is described in a 1970 monograph of Foata and Schützenberger.

Recall the recurrence we obtained for the Stirling numbers back in Problem 2.23. For integers $n \geq 1$ and $1 \leq k \leq n$, we have

$$\left\{ n \atop k \right\} = \left\{ n-1 \atop k-1 \right\} + k \left\{ n-1 \atop k \right\}.$$

We also encountered a recurrence for the binomial coefficients (called Pascal's Recurrence) in Problem 2.33. For integers $n \geq 1$ and $0 \leq k \leq n$, we have

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Our next goal is to obtain a similar recurrence for the Eulerian numbers.

For $n \in \mathbb{N}$, fix $w \in S[n]$ with k descents. If we consider the one-line notation for w, we can obtain a permutation $\hat{w} \in S[n-1]$ by deleting n from w's one-line notation. For example, if $w = 3142 \in S[4]$, then $\hat{w} = 312 \in S[3]$. If w has k descents, then $\hat{w}$ has either k or $k-1$ descents. Conversely, we can construct permutations in $S[n]$ with k descents from permutations in $S[n-1]$ with k or $k-1$ descents by inserting n in the one-line notation. Flesh out these ideas to tackle the next problem.

Problem 4.14. Prove that for integers $n \geq 1$ and $0 \leq k \leq n-1$, we have

$$\left\langle n \atop k \right\rangle = (n-k) \left\langle n-1 \atop k-1 \right\rangle + (k+1) \left\langle n-1 \atop k \right\rangle.$$

Problem 4.15. Confirm that

$$\left\langle 4 \atop 1 \right\rangle = 3 \left\langle 3 \atop 0 \right\rangle + 2 \left\langle 3 \atop 1 \right\rangle$$

by looking at Table 4.2.

For $n \geq 1$, we define the nth **Eulerian polynomial**, denoted $S_n(t)$, as the (ordinary) generating function for the Eulerian numbers $\left\langle n \atop k \right\rangle$ with $0 \leq k \leq n-1$. That is, we have

$$S_n(t) := \sum_{w \in S[n]} t^{\mathrm{des}(w)} = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle t^k.$$

Take a moment to convince yourself that the two sums above are equal. We define $S_0(t) := 1$, since $S[0]$ contains only the empty permutation, which has no descents.

Problem 4.16. Compute $S_0(t)$, $S_1(t)$, $S_2(t)$, $S_3(t)$, $S_4(t)$, and $S_5(t)$ using Table 4.2.

Problem 4.17. For $n \in \mathbb{N} \cup \{0\}$, determine the value of each of the following.

(a) $S_n(0)$

(b) $S_n(1)$

It turns out that there is reasonably nice linear recurrence for the Eulerian polynomials.

Problem 4.18. For $n \in \mathbb{N} \cup \{0\}$, prove that

$$S_{n+1}(t) = (1 + nt)S_n(t) + t(1 - t)S_n'(t).$$

Hint: This is an exercise in manipulatorics! Fiddle around with the righthand side of equation above until you obtain the lefthand side. Along the way, you will need to shift some indices and cleverly add zero in a couple spots. For example:

$$(n + 1)\left\langle {n \atop n} \right\rangle t^n = 0 \quad \text{and} \quad (n - 0 + 1)\left\langle {n \atop -1} \right\rangle t^0 = 0.$$

Problem 4.19. We define a **barred permutation** as follows. Given $w \in S[n]$ with $n \in \mathbb{N} \cup \{0\}$, we must place at least one vertical bar after each descent position and we can place finitely many additional vertical bars in gaps that do not correspond to descents or on the ends. For example, $1|5|237||46|$ is a barred permutation with 5 vertical bars. A single vertical bar was required after 5 and 7, respectively, and the other three bars were optional.

(a) Explain why there is a bijection between the collection of barred permutations on n with k bars and the collection of configurations of n labeled balls into $k + 1$ labeled buckets (see Problem 2.10).

Hint: As an example, the barred permutation $1|5|237||46|$ corresponds to placing 7 balls into 6 buckets such that the first bucket contains ball 1, the second bucket contains ball 5, the third bucket contains balls 2, 3, 7, the fourth bucket is empty, the fifth bucket has balls 4 and 6, and the sixth bucket is empty. The underlying permutation for this example is $w = 1523746$.

(b) Prove that the (ordinary) generating function for the number of barred permutations that have a fixed $w \in S[n]$ as their underlying permutation is given by

$$\frac{t^{\mathrm{des}(w)}}{(1 - t)^{n+1}}.$$

Hint: Utilize a factor of $1 + t + t^2 + \cdots$ when there is no descent and a factor of $t + t^2 + t^3 + \cdots$ if there is a descent.

(c) Explain why the (ordinary) generating function for the number of ways to put n labeled balls into $k + 1$ labeled buckets is

$$\sum_{k \geq 0} (k + 1)^n t^k.$$

(d) Prove that

$$\frac{S_n(t)}{(1-t)^{n+1}} = \sum_{k\geq 0}(k+1)^n t^k.$$

This last equation is referred to as the **Carlitz Identity**.

Problem. Let $n, i \geq 0$ be integers. Prove that

$$\frac{t^i}{(1-t)^{n+1}} = \sum_{k\geq 0}\binom{n+k-i}{n}t^k.$$

Hint: Use Problems 3.18 and 3.19 to expand $1/(1-t)^{n+1}$, and apply binomial symmetry from Problem 2.32. Then multiply by t^i and reindex. Why are the terms with $k < i$ zero?

Problem 4.20. Let $n \geq 1$ and $k \geq 0$ be integers. Prove that

$$(k+1)^n = \sum_{i=0}^{n-1}\left\langle {n \atop i} \right\rangle \binom{k+n-i}{n}.$$

This is referred to as **Worpitzky's Identity**.

Hint: Use the preceding problem to help rewrite the lefthand side of the Carlitz Identity as a double sum, and then equate coefficients.

Problem 4.21. Verify Worpitzky's Identity holds for $n = 5$ and $k = 3$.

We can use Worpitzky's Identity repeatedly to get explicit formulas for the Eulerian numbers.

Problem 4.22. Use Worpitzky's Identity to verify the following known results.

(a) $\left\langle {n \atop 0} \right\rangle = 1$

(b) $\left\langle {n \atop 1} \right\rangle = 2^n - n - 1$

Problem 4.23. Use Worpitzky's Identity to obtain a closed form expression for $\left\langle {n \atop 2} \right\rangle$.

One can generalize the approach implemented above to obtain a formula for computing an Eulerian number. The resulting formula is provided in the theorem below. We will not take the time to prove this formula, but if one were so inclined, one approach is to multiply the Carlitz Identity from Problem 4.19 by $(1-t)^{n+1}$, expand this factor using the Binomial Theorem, and compare coefficients of t^k. Alternatively, the formula can be verified by induction on n using the recurrence for the Eulerian numbers.

Theorem 4.24 (Alternating Sum Formula for Eulerian Numbers)**.** For integers $n \geq 1$ and $0 \leq k \leq n-1$, we have

$$\left\langle {n \atop k} \right\rangle = \sum_{i=0}^{k}(-1)^i\binom{n+1}{i}(k+1-i)^n.$$

The goal for the rest of this section is to produce an exponential generating function for the Eulerian polynomials. Let $w \in S[n]$ and assume that $w(i+1) = n$. Figure 4.2 provides a representation of the graph of w. Inside the dotted box on the left is a permutation of i values while inside the dotted box on the right, we have a permutation of $n-1-i$ values, where the two sets of values are disjoint. Note that it is possible that one of the dotted boxes (or both if $n = 1$) is empty. There are $\binom{n-1}{i}$ many ways to choose the values to be used in the left box, which then forces the remaining values from $[n-1]$ to be utilized in the right box. Given a subset of i values to be used in the left box, the descent distribution for all possible permutations in that box is determined by the Eulerian polynomial $S_i(t)$. Similarly, the descent distribution for all possible permutations in the right box is determined by the Eulerian polynomial $S_{n-1-i}(t)$.

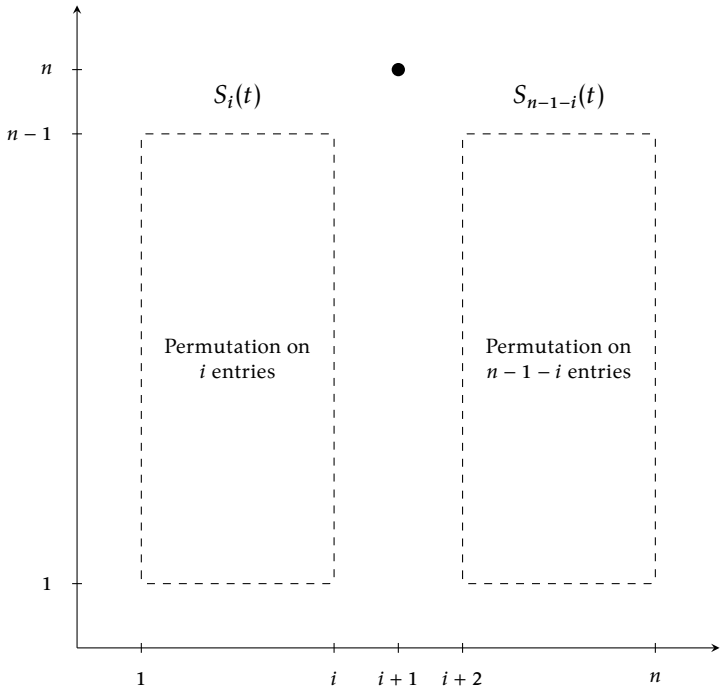


Figure 4.2: Splitting a permutation at its largest entry in position $i+1$. The dashed regions contain the entries to its left and right.

Problem 4.25. Let $n \geq 1$ and $0 \leq i \leq n-1$ be integers. Fix $L \subseteq [n-1]$ with $|L| = i$, let $R = [n-1] \setminus L$, and define the set

$$S[n]_L^{i+1} := \{w \in S[n] \mid w(i+1) = n \text{ and } w(1), \ldots, w(i) \in L\}.$$

Notice that every $w \in S[n]_L^{i+1}$ has one-line notation in the form $u \cdot n \cdot v$, where u is a linear order of the elements in L, v is a linear order of the elements in R, and we interpret $u \cdot n \cdot v$ as concatenation of u, n, and v. For example, if $w = 3172564 \in S[7]$, then $i+1 = 3$ since the maximum value of w appears in position 3. Here, $u = 31$ and $v = 2564$ with $L = \{1, 3\}$ and $R = \{2, 4, 5, 6\}$.

(a) If $i+1 < n$ and $w \in S[n]_L^{i+1}$ has one-line notation $u \cdot n \cdot v$, then explain why $\operatorname{des}(w) = \operatorname{des}(u) + 1 + \operatorname{des}(v)$.

(b) If $i+1 < n$, then explain why the generating function for descents of permutations in $S[n]_L^{i+1}$ is given by

$$S_i(t) \cdot t \cdot S_{n-1-i}(t).$$

(c) If $i+1 = n$, then explain why the generating function for descents of permutations in $S[n]_L^n$ is given by $S_{n-1}(t)$.

(d) Explain why

$$S_n(t) = S_{n-1}(t) + t \sum_{i=0}^{n-2} \binom{n-1}{i} S_i(t) S_{n-1-i}(t),$$

where the sum is interpreted as zero when $n = 1$. The equation above is a quadratic recurrence for the Eulerian polynomials.

Define the exponential generating series for the Eulerian polynomials via

$$S(t,z) := \sum_{n \geq 0} S_n(t) \frac{z^n}{n!} = \sum_{n,k \geq 0} \left\langle {n \atop k} \right\rangle t^k \frac{z^n}{n!}.$$

Take a moment to let this sink in. Above, we have a bivariate generating series. Viewed as an exponential generating series in z, it has the Eulerian polynomial $S_n(t)$ as the coefficient of $z^n/n!$.

Our next goal is to obtain a closed form for $S(t,z)$. The method of attack is to produce a differential equation that we can solve. If we differentiate with respect to z, treating t as a fixed parameter, we obtain

$$\begin{aligned}
\frac{\partial S}{\partial z}(t,z) &= \sum_{n \geq 1} S_n(t) \frac{z^{n-1}}{(n-1)!} \\
&= \sum_{n \geq 1} S_{n-1}(t) \frac{z^{n-1}}{(n-1)!} + t \sum_{n \geq 1} \sum_{i=0}^{n-2} \binom{n-1}{i} S_i(t) S_{n-1-i}(t) \frac{z^{n-1}}{(n-1)!} \\
&= S(t,z) + t \sum_{n \geq 1} \sum_{i=0}^{n-2} S_i(t) \frac{z^i}{i!} S_{n-1-i}(t) \frac{z^{n-1-i}}{(n-1-i)!} \\
&= S(t,z) + t \left(\sum_{i \geq 0} S_i(t) \frac{z^i}{i!} \right) \left(\sum_{j \geq 1} S_j(t) \frac{z^j}{j!} \right) \\
&= S(t,z) + t S(t,z) \big(S(t,z) - 1 \big) \\
&= (1-t) S(t,z) + t S(t,z)^2.
\end{aligned}$$

The second equality uses the quadratic recurrence from Problem 4.25. To factor the double sum, we set $j = n-1-i$. Since $0 \leq i \leq n-2$, we have $j \geq 1$, so the second factor omits the constant term $S_0(t) = 1$. Thus, for fixed t, the series $S(t,z)$ satisfies the differential equation

$$\frac{\partial}{\partial z} S(t,z) = (1-t) S(t,z) + t S(t,z)^2 \text{ with } S(t,0) = 1.$$

If we solve this differential equation with the initial condition, we get the following theorem, which is due to…you guessed it, Euler.

Theorem 4.26. The exponential generating function for the Eulerian polynomials satisfies

$$S(t,z) = \frac{t-1}{t-e^{z(t-1)}}.$$

Let's check that this closed form recovers two counts we already know. Setting $t = 1$ ignores the number of descents, so we expect to recover the exponential generating function for the sequence determined by $n!$ for $n \geq 0$. Setting $t = 0$ selects permutations with no descents, so we expect to recover the exponential generating function for the constant sequence $1, 1, \ldots$.

First, substituting $t = 1$ into the quotient gives $0/0$, so we cannot evaluate it directly at $t = 1$. However, since $S_n(1) = n!$, the generating series itself gives

$$S(1,z) = \sum_{n\geq 0} S_n(1)\frac{z^n}{n!} = \sum_{n\geq 0} z^n = \frac{1}{1-z},$$

which is the exponential generating function for the sequence determined by $n!$, as expected. Thus, we interpret the closed form at $t = 1$ by its limiting value. For $z \neq 1$, L'Hôpital's Rule confirms that

$$\lim_{t\to 1} \frac{t-1}{t-e^{z(t-1)}} = \lim_{t\to 1} \frac{1}{1-ze^{z(t-1)}} = \frac{1}{1-z}.$$

Second, substituting $t = 0$ into the closed form gives $S(0,z) = e^z$, which is the exponential generating function for the constant sequence $1, 1, \ldots$, as expected. For each $n \geq 0$, there is exactly one permutation in $S[n]$ with no descents, including the empty permutation when $n = 0$.

4.2 Catalan Numbers

For $n \in \mathbb{N} \cup \{0\}$, a permutation $w \in S[n]$ is said to **contain the pattern** 231 if there exist indices (positions) $1 \leq i < j < k \leq n$ such that

$$w(k) < w(i) < w(j).$$

Notice that indices i, j, k need not be consecutive. The pattern 231 captures the relative order "medium, big, small" of values from left to right in the one-line notation of a permutation. A permutation of length less than three, including the empty permutation, cannot contain this pattern because there are not enough indices.

Example 4.27. Consider the permutation $w = 53412 \in S[5]$. We see that w contains the pattern 231 twice: once in positions 2, 3, 4 with values 3, 4, 1, respectively, and again in positions 2, 3, 5 with values 3, 4, 2, respectively. We have highlighted the two instances of the pattern in Figure 4.3.

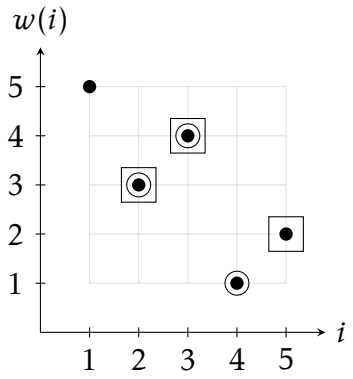


Figure 4.3: The graph of $w = 53412$ from Example 4.27. Circles mark the occurrence of 231 with values 3, 4, 1; squares mark the occurrence with values 3, 4, 2.

Problem 4.28. Find all permutations in $S[4]$ that contain at least one instance of the pattern 231.

If a permutation $w \in S[n]$ does not contain the pattern 231, then we say that w is **231-avoiding**, and we define

$$S[n](231) := \{w \in S[n] \mid w \text{ is 231-avoiding}\}.$$

Example 4.29. If we organize the permutations in $S[4](231)$ according to the number of descents, we obtain the distribution in Table 4.4. If we count the total number of permutations in the table, we see that $|S[4](231)| = 14$.

$\mathrm{des}(w) = 0$	$\mathrm{des}(w) = 1$	$\mathrm{des}(w) = 2$	$\mathrm{des}(w) = 3$
1234	1243	1432	4321
	1324	2143	
	1423	3214	
	2134	4132	
	3124	4213	
	4123	4312	

Table 4.4: Permutations in $S[4](231)$ organized by number of descents.

Problem 4.30. For $n \in \{0, 1, 2, 3\}$, determine $S[n](231)$.

For $n \in \mathbb{N} \cup \{0\}$, we define the **$n$th Catalan number** via

$$c_n := |S[n](231)|.$$

Problem 4.31. Determine c_0, c_1, c_2, c_3, and c_4.

The next problem will walk us through the construction of the **Catalan recurrence**.

Problem 4.32. Let $n \geq 1$.

(a) Fix $w \in S[n](231)$ and assume $w(i+1) = n$ for $1 \leq i+1 \leq n$. Figure 4.4 depicts the structure of the graph of such a w, where each of the dashed boxes may be empty. Explain why the dashed boxes must have the relative heights as shown in the figure.

(b) For each $0 \le i \le n-1$, explain why there is a bijection between permutations $w \in S[n](231)$ satisfying $w(i+1) = n$ and pairs $(u, v) \in S[i](231) \times S[n-1-i](231)$. Then explain why the number of such permutations is $c_i c_{n-1-i}$.

(c) Explain why

$$c_n = \sum_{i=0}^{n-1} c_i c_{n-1-i}.$$

(d) Use the initial condition $c_0 = 1$ and the recurrence to find c_5.

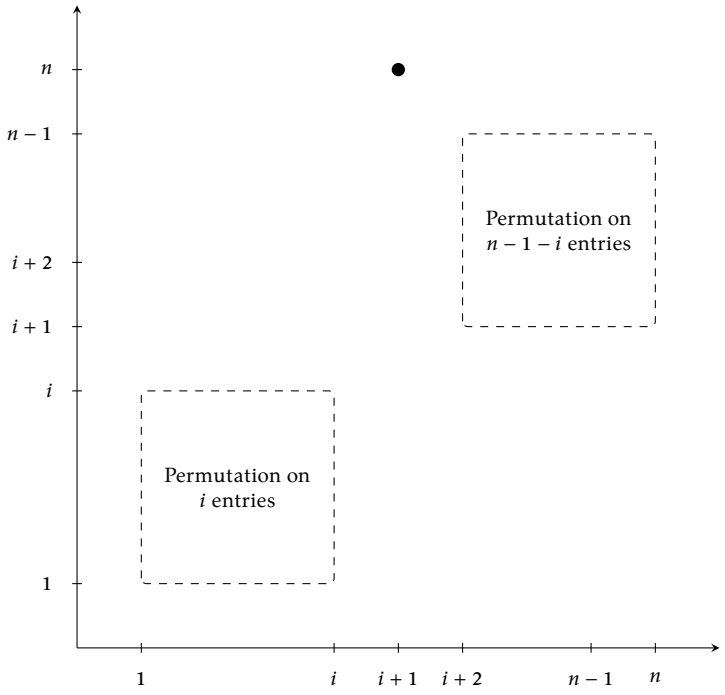


Figure 4.4: Splitting a 231-avoiding permutation at its largest entry in position $i+1$. The entries to its left lie below all entries to its right. The picture shows the case where both sides are nonempty.

The Catalan numbers are uniquely determined by their initial value and recurrence. Indeed, if a sequence $(a_n)_{n\ge 0}$ satisfies $a_0 = 1$ and

$$a_n = \sum_{i=0}^{n-1} a_i a_{n-1-i} \qquad (n \ge 1),$$

then induction implies that $a_n = c_n$ for every $n \ge 0$.

There are hundreds of interesting combinatorial objects counted by the Catalan numbers! Let's explore a few.

A **planar binary tree** is a finite rooted tree in which every node has either no children or exactly two children, distinguished as its left child and its right child. A node with

two children is called an **internal node**, and a node with no children is called a **leaf**. The nodes are unlabeled, and only the branching structure and the left-to-right order matter, not the lengths or angles of the edges. In particular, exchanging the left and right subtrees can produce a different planar binary tree.

For $n \in \mathbb{N} \cup \{0\}$, let $\mathrm{PB}[n]$ denote the set of planar binary trees with n internal nodes, and let $\mathrm{pb}_n := |\mathrm{PB}[n]|$. The set $\mathrm{PB}[0]$ contains a single tree consisting of one leaf, so $\mathrm{pb}_0 = 1$. Figure 4.5 shows all five trees in $\mathrm{PB}[3]$, so $\mathrm{pb}_3 = 5$. In the figure, internal nodes are drawn in black and leaves are drawn in white.

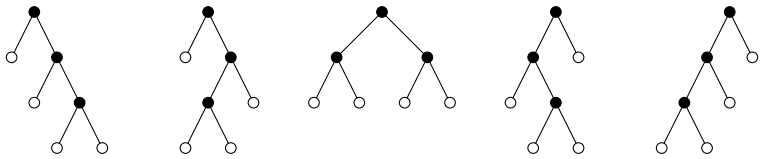


Figure 4.5: The five planar binary trees with three internal nodes. Internal nodes are black, and leaves are white.

Problem 4.33. Let $n \in \mathbb{N} \cup \{0\}$.

(a) Explain why a planar binary tree with n internal nodes has $n + 1$ leaves.

(b) Prove that planar binary trees are enumerated by the Catalan numbers. That is, show that $\mathrm{pb}_n = c_n$ for every $n \geq 0$.

Hint: Verify the initial value. Then split a tree with $n \geq 1$ internal nodes at its root and verify the Catalan recurrence.

In light of the previous problem, $|\mathrm{PB}[n]| = |S[n](231)|$. Let's describe a bijection $f : \mathrm{PB}[n] \to S[n](231)$ that constructs a permutation in one-line notation directly from a tree. Begin with a tree $T \in \mathrm{PB}[n]$ and the available values $1, \ldots, n$. More generally, the recursive procedure takes a subtree and a set of available values, one for each of its internal nodes. If the subtree consists of a single leaf, its set of available values is empty and it contributes the empty word. Otherwise, proceed as follows:

(a) Reserve the largest available value for the position between the left and right subtree words.

(b) If the left subtree has ℓ internal nodes, give it the ℓ smallest remaining values and give the right subtree the rest.

(c) Apply the same procedure to each subtree using its assigned values. Concatenate the left subtree word, the reserved value, and the right subtree word, in that order.

For $n \geq 1$, at the root of T this places n in position $\ell + 1$, with ℓ entries to its left and $n - 1 - \ell$ entries to its right. At each stage, only the relative order of the available values matters; we use those values directly without renumbering them. In particular, a tree with one internal node produces the word 1.

At each stage, the word we construct has the form illustrated in Figure 4.4: its largest entry separates a left word whose entries are all smaller than those in the right word. Applying this rule recursively ensures that each subtree word avoids 231, while the relative sizes of the entries prevent a 231 pattern from involving more than one of the three parts. Thus the construction produces a 231-avoiding permutation. Conversely, split a 231-avoiding permutation at its largest entry n. Every entry to the left of n must be smaller than every entry to its right; otherwise, those two entries together with n would form a 231 pattern. If there are ℓ entries to the left of n, those entries must be precisely the values $1, \dots, \ell$. Recursively split each remaining word at its largest entry to recover the left and right subtrees, using the values already present in each word. An empty word corresponds to a leaf. This reverses the construction and shows that f is a bijection.

Example 4.34. Let $T \in \mathrm{PB}[9]$ be the tree in Figure 4.6. Its left subtree L has five internal nodes, and its right subtree R has three. Thus 9 occupies position 6, while L receives the values $1, \dots, 5$ and R receives $6, 7, 8$. Blanks indicate positions whose entries have not yet been determined:

$$f(T) = \underbrace{_\ _\ _\ _\ _}_{1\text{–}5}\ 9\ \underbrace{_\ _\ _}_{6\text{–}8}.$$

The root of L has one internal node in its left subtree and three in its right subtree. Reserve 5, give 1 to the left subtree, and give $2, 3, 4$ to the right subtree:

$$f(T) = \underbrace{_}_{1}\ 5\ \underbrace{_\ _\ _}_{2\text{–}4}\ 9\ \underbrace{_\ _\ _}_{6\text{–}8}.$$

In the latter subtree, reserve 4, give 2 to its left subtree, and give 3 to its right subtree:

$$f(T) = \underbrace{_}_{1}\ 5\ \underbrace{_}_{2}\ 4\ \underbrace{_}_{3}\ 9\ \underbrace{_\ _\ _}_{6\text{–}8}.$$

Each of the subtrees receiving a single value has one internal node, so their words are 1, 2, and 3, respectively. Consequently, the left subtree word is 15243, so we have

$$f(T) = 1\ 5\ 2\ 4\ 3\ 9\ \underbrace{_\ _\ _}_{6\text{–}8}.$$

The root of R has two internal nodes in its left subtree and a leaf on its right:

$$f(T) = 1\ 5\ 2\ 4\ 3\ 9\ \underbrace{_\ _}_{6\text{–}7}\ 8.$$

Reserve 8 and give $6, 7$ to the left subtree. The root of this subtree has a leaf on the left and one internal node on the right, so its word is 76. Thus the word for R is 768. Combining the two subtree words gives

$$f(T) = 1\ 5\ 2\ 4\ 3\ 9\ 7\ 6\ 8.$$

Problem 4.35. Let $T \in \mathrm{PB}[9]$ be the planar binary tree in Figure 4.7. Find $f(T)$ in one-line notation, where $f : \mathrm{PB}[n] \to S[n](231)$ is the map described above.

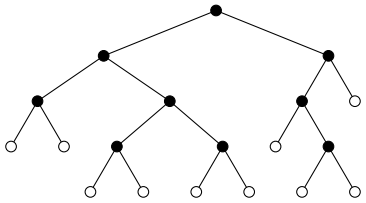


Figure 4.6: The planar binary tree $T \in \mathrm{PB}[9]$ corresponding to $f(T) = 152439768$. Internal nodes are black, and leaves are white.

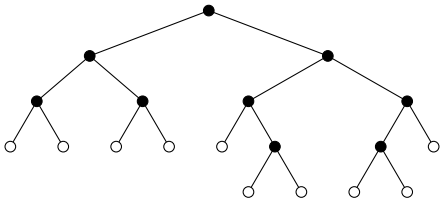


Figure 4.7: The planar binary tree $T \in \mathrm{PB}[9]$ for Problem 4.35. Internal nodes are black, and leaves are white.

For $n \geq 0$, a **Dyck path** of **semilength** n is a path in the plane that starts at $(0, 0)$, takes n steps one unit North and n steps one unit East, and ends at (n, n) without ever going below the line $y = x$. Let $\mathrm{Dyck}[n]$ denote the set of these paths and let $d_n := |\mathrm{Dyck}[n]|$. The unique path with no steps gives $d_0 = 1$. Figure 4.8 shows a Dyck path of semilength 5.

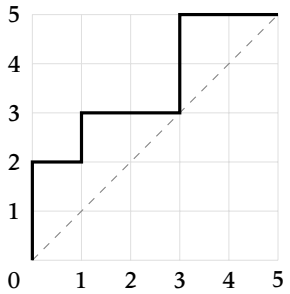


Figure 4.8: A Dyck path in $\mathrm{Dyck}[5]$. The dashed line is $y = x$.

Problem 4.36. Let's explore Dyck paths.

(a) Draw all paths in $\mathrm{Dyck}[n]$ for $1 \leq n \leq 4$ and determine their counts.

(b) For $n \geq 1$, prove that

$$d_n = \sum_{i=0}^{n-1} d_i d_{n-1-i}.$$

We can conclude that $d_n = c_n$ for every $n \geq 0$.

Hint: Split at the first return to $y = x$, say $(i + 1, i + 1)$, as illustrated in Figure 4.9. Think about how many ways you can draw the Dyck path to get to $(i+1, i+1)$ versus

how many ways you can draw the Dyck path from $(i+1, i+1)$ to (n, n). The first case is trickier to think about. Notice that the portion of the Dyck path from $(0, 0)$ to $(i+1, i+1)$ never hits the line $y = x$ along the way. Moreover, this portion necessarily starts with a North step and ends with an East step. What are the possible values for i?

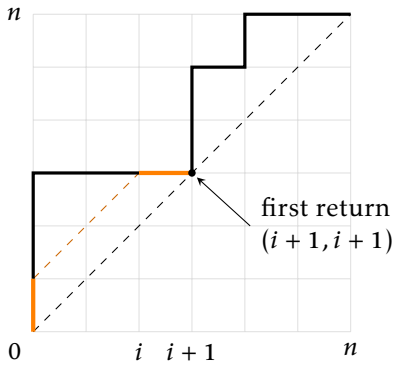


Figure 4.9: Decomposition at the first return to $y = x$. The highlighted North and East steps enclose a translated Dyck path of semilength i; the remaining path has semilength $n - 1 - i$. The black portions illustrate possible choices.

(c) For $n \geq 0$, show that there are $\binom{2n}{n}$ lattice paths from $(0, 0)$ to (n, n) using only steps one unit North or one unit East. These paths may pass below $y = x$.

(d) For $n \geq 1$, show that there are $\binom{2n}{n-1}$ lattice paths from $(0, 0)$ to $(n+1, n-1)$ using only steps one unit North or one unit East.

(e) For $n \geq 1$, construct a bijection between the lattice paths from $(0, 0)$ to (n, n) that pass below $y = x$ and the lattice paths from $(0, 0)$ to $(n+1, n-1)$. All steps are one unit North or one unit East.

Hint: Find the first vertex below $y = x$, as illustrated in Figure 4.10. Reflect the portion of the path after this vertex across the line $y = x - 1$. Explain how to reverse this construction.

(f) Deduce that, for $n \geq 1$,

$$c_n = d_n = \binom{2n}{n} - \binom{2n}{n-1}.$$

One can simplify the expression above to obtain

$$c_n = \frac{1}{n+1}\binom{2n}{n}.$$

This formula also holds for $n = 0$.

Now, define the (ordinary) generating series for the Catalan numbers via

$$C(t) := \sum_{n \geq 0} c_n t^n.$$

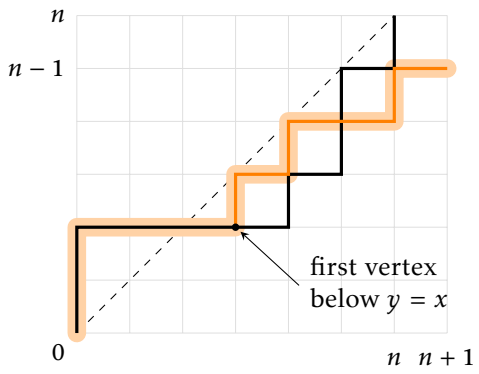


Figure 4.10: A lattice path from $(0,0)$ to (n,n) that passes below $y = x$, shown for $n = 6$. The marked vertex is the endpoint of the first step below the diagonal. The orange path is the reflection of the continuation after this vertex across $y = x-1$ and ends at $(n+1, n-1)$. The highlight traces the resulting path from $(0,0)$ to $(n+1, n-1)$.

Problem 4.37. Let's see if we can find a closed form for $C(t)$.

(a) Prove that $C(t) = 1 + t \cdot C(t)^2$.

Hint: Isolate the constant term, rewrite c_n using the recurrence for $n \geq 1$ to obtain a double sum, factor out t, distribute the remaining $n-1$ factors of t in the double sum to c_i and c_{n-1-i}, and then rewrite the double sum as a product of two sums.

(b) Using part (a), prove that

$$C(t) = \frac{1 - \sqrt{1-4t}}{2t}.$$

Along the way, you will be confronted with two possible solutions arising from using the quadratic formula. Justify that we are using the correct option.

The displayed expression is undefined at $t = 0$, but we can extend it to $t = 0$ by assigning it the value of its right-hand limit as $t \to 0^+$. Verify that this limit is 1, agreeing with $C(0) = c_0 = 1$.

(c) Using the Maclaurin series for $C(t)$, find c_6. You are welcome to use technology to find the Maclaurin series.

We now have four ways to prove that a family of finite sets $\{A_n\}_{n\geq 0}$ is enumerated by the Catalan numbers, meaning that $|A_n| = c_n$ for every $n \geq 0$:

(i) Verify that the numbers $|A_n|$ satisfy the initial condition and Catalan recurrence.

(ii) For each $n \geq 0$, construct a bijection between A_n and a set known to have c_n elements.

(iii) Prove directly that $|A_n| = \frac{1}{n+1}\binom{2n}{n}$ for every $n \geq 0$, using the formula established in Problem 4.36.

(iv) Verify that the ordinary generating function for the numbers $|A_n|$ has the closed form given in Problem 4.37.

Problem 4.38. A **balanced sequence of parentheses** has equally many left and right parentheses, and every prefix has at least as many left as right parentheses. For example, the sequences with three pairs are

$$()()(), \qquad ()(()), \qquad (()()), \qquad (())(), \qquad ((())).$$

Prove that there are c_n balanced sequences of parentheses with n pairs, including the empty sequence when $n = 0$.

Problem 4.39. For $n \geq 1$, a **triangulation** divides a fixed convex $(n+2)$-gon into triangles using noncrossing diagonals. Rotations and reflections are not identified. Figure 4.11 shows all five triangulations of a fixed convex pentagon. Prove that there are c_n triangulations of a convex $(n+2)$-gon.

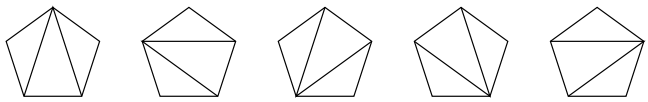


Figure 4.11: The five triangulations of a fixed convex pentagon, illustrating $c_3 = 5$.

Problem 4.40. A **planar rooted tree** is a finite rooted tree in which each node may have any number of children, ordered from left to right. The nodes are unlabeled, and only the branching structure and the order of the children matter. Unlike our planar binary trees, these trees need not have exactly two children at each internal node. We do not include an extra edge below the root (a "trunk"). For $n \geq 0$, let $\mathrm{PRT}[n]$ denote the set of planar rooted trees with n edges and let $\mathrm{prt}_n := |\mathrm{PRT}[n]|$. In particular, $\mathrm{PRT}[0]$ consists of the single-node tree, so $\mathrm{prt}_0 = 1$. Figure 4.12 shows a tree in $\mathrm{PRT}[11]$. Determine prt_n.

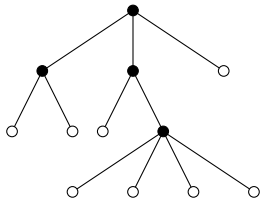


Figure 4.12: A planar rooted tree with eleven edges. Internal nodes are black and leaves are white.

For the next problem, extend our definition of 231-avoidance to each of the other patterns of length three: an occurrence of $p \in S[3]$ consists of three entries, read from left to right, whose relative order is p. Write $S[n](p)$ for the set of permutations in $S[n]$ that avoid p.

The following map is a useful bijection. For $n \geq 0$, define the **complement map** $\mathrm{comp} : S[n] \to S[n]$ via

$$\mathrm{comp}(w) = \big(n+1-w(1)\big)\cdots\big(n+1-w(n)\big)$$

in one-line notation, where the parenthesized expressions are individual entries, not factors in a product. This map reflects the graph of w across the horizontal midline $y = (n+1)/2$. Both reverse and complement are involutions, and they commute.

For the next problem, consider how the reverse map, the complement map, and their composition change occurrences of patterns. These maps connect the four avoidance sets appearing in the problem.

Problem 4.41. Prove that

$$|S[n](p)| = c_n$$

for every $n \geq 0$ and every $p \in \{132, 213, 231, 312\}$.

We will handle the remaining patterns, 123 and 321, in Section 4.3, where a bijection between 321-avoiding permutations and Dyck paths will also connect permutation statistics with path statistics.

4.3 Narayana Numbers

More coming soon…